

Mechanical Technology

Welding

Joining methods

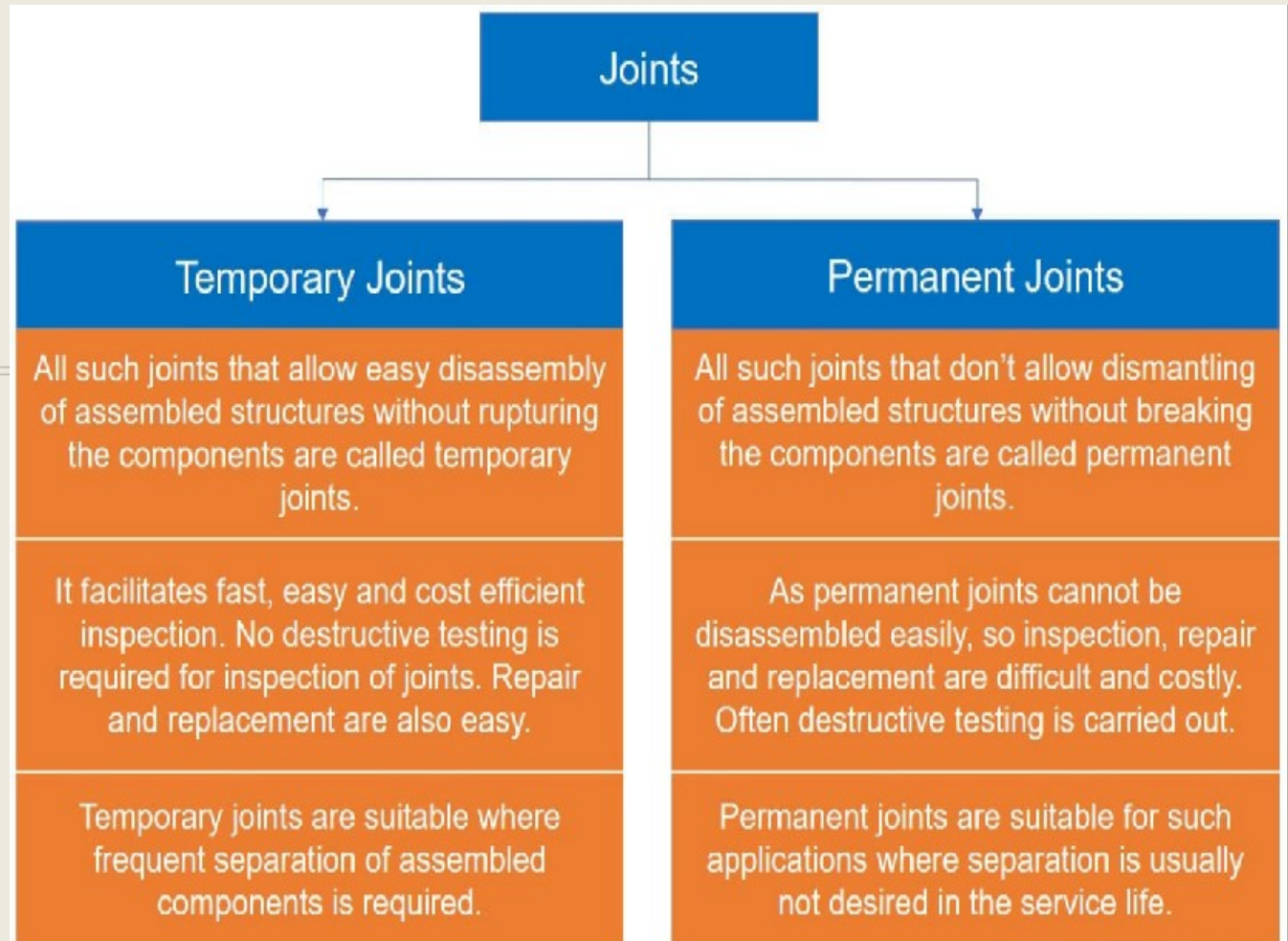
By the end of the topic learners will be able to:-

Explain the following types of joining methods:

- ☐ Riveting
- ☐ Use nuts and bolts
- ☐ Welding
- ☐ Prepare material before a tack weld
- ☐ Perform a tack weld
- ☐ Perform a butt weld no longer than 30 mm
- ☐ Clean and present a butt weld

Joining Processes

Joining Processes are the processes of welding, riveting, bolting and soldering parts together after it has been manufactured. It is one of the final processes in manufacturing and also one of the most important processes.

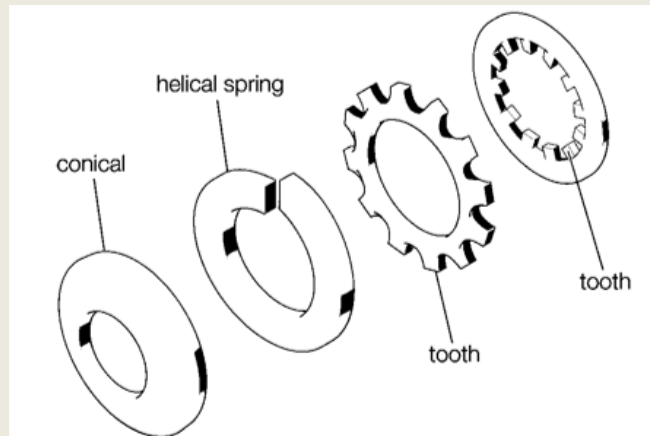
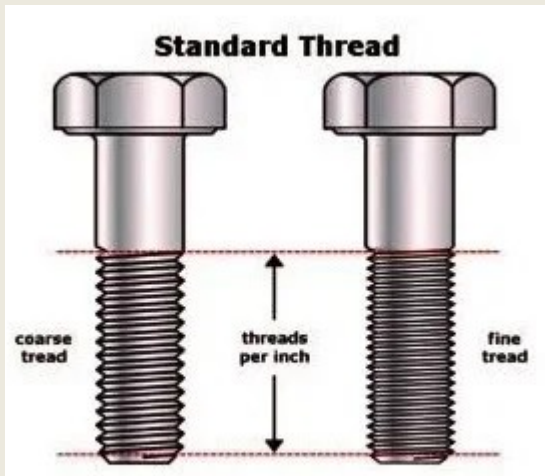


Temporary Jointing Methods

A nut is a small metal object with a hole in the middle that has a corrugated hole. These curved holes are known as threads. Nuts are used as a fastening device.

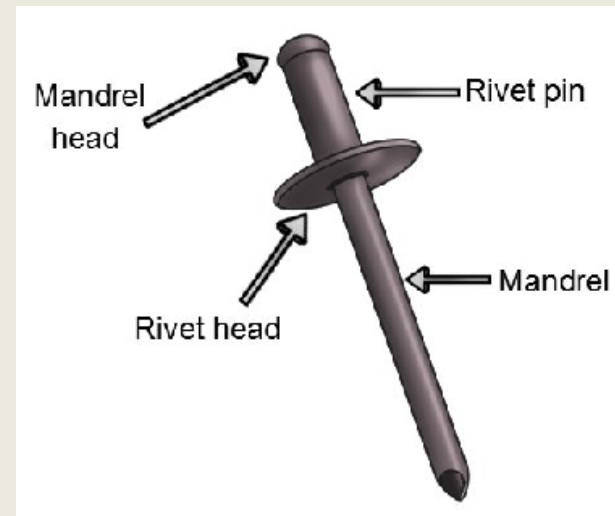
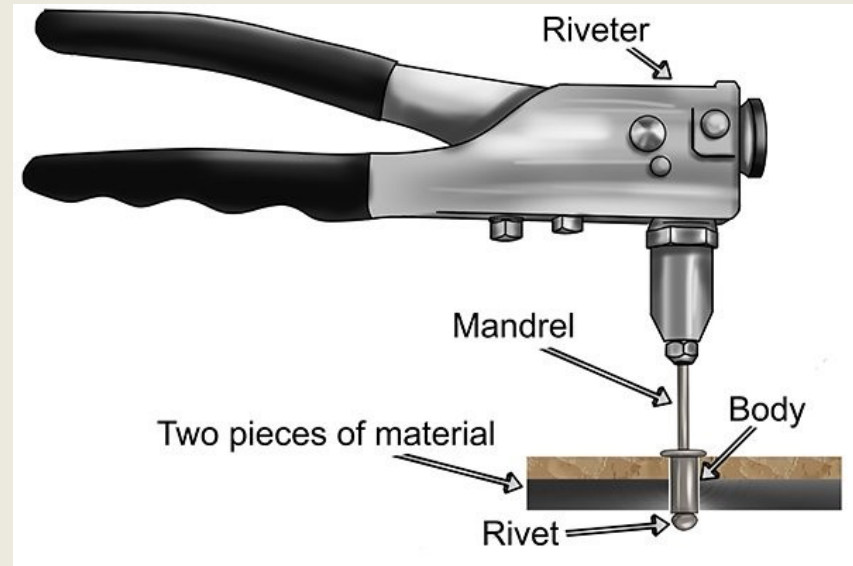
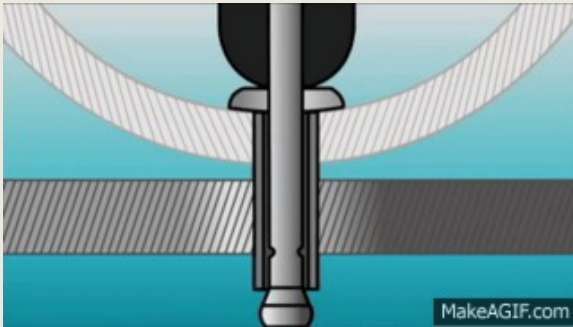
Bolts are metal objects consisting of a cylindrical trunk with grooves such as grooves that are similar to the grooves present inside a nut.

The primary purpose of most washers is to evenly distribute the load of the threaded fastener with which they are used.



Permanent Jointing Methods

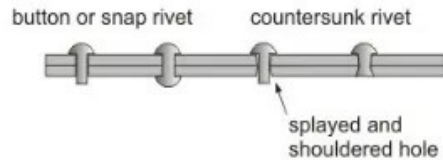
- In certain instances, permanent joining of materials are preferred. Especially where structural strength and permanent shape is required.



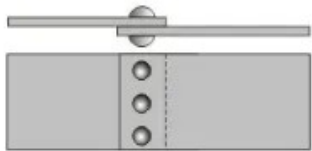
Mar
he

Traditional Rivets

Rivets



single riveted lap joint



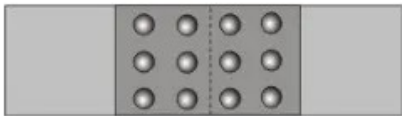
single fish plate joint - one cover plate



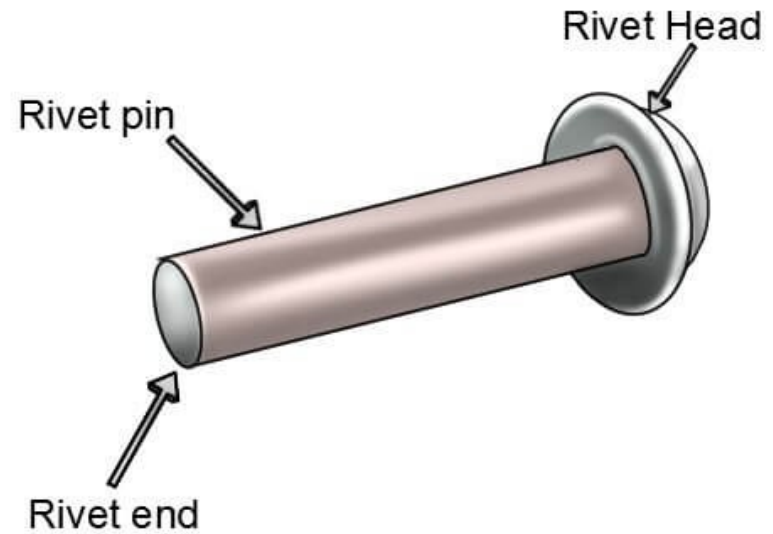
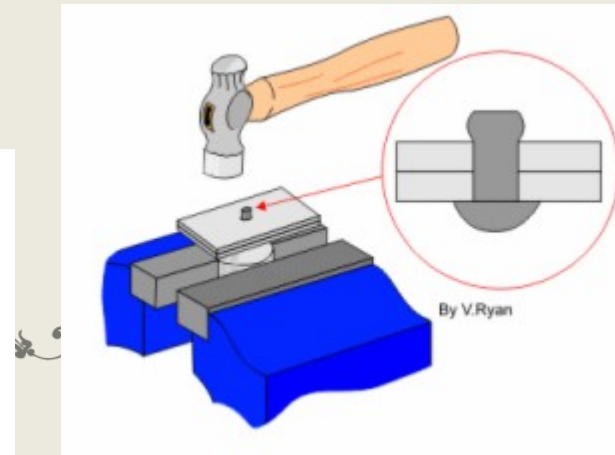
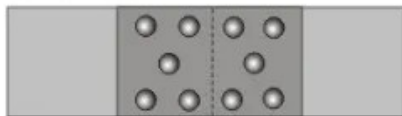
two cover plates



double riveted joints - chain pattern

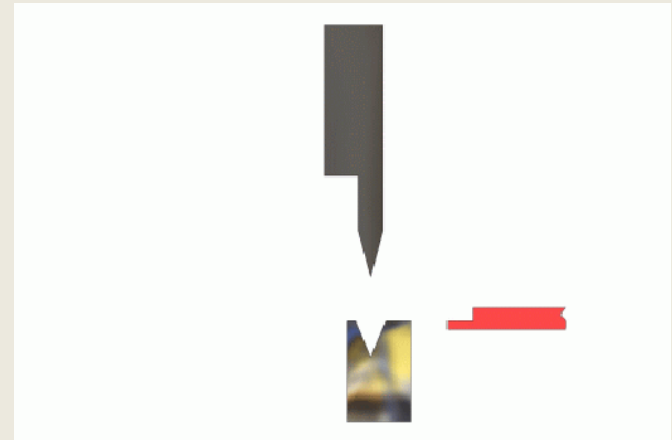
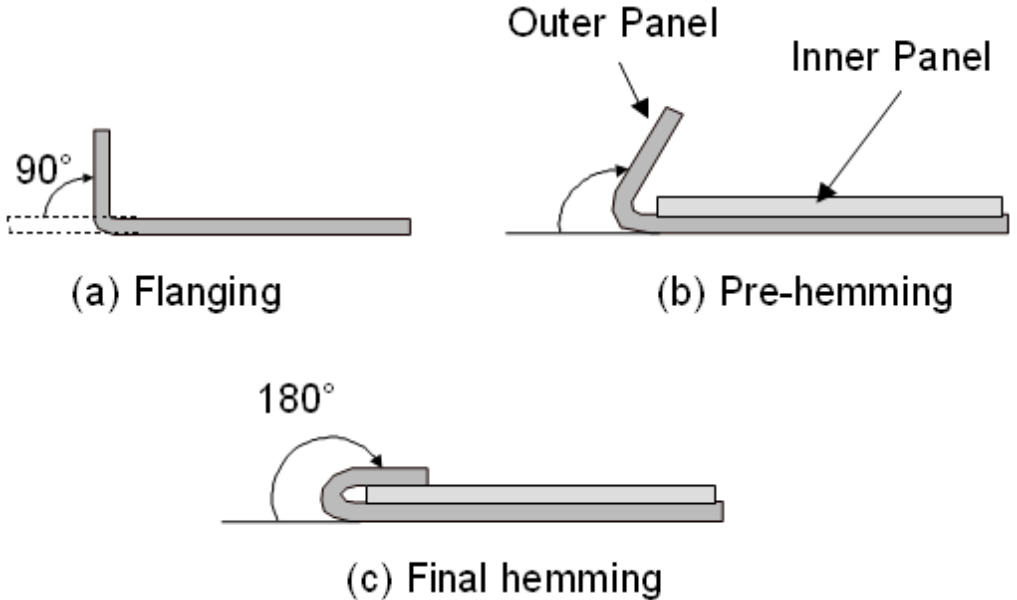
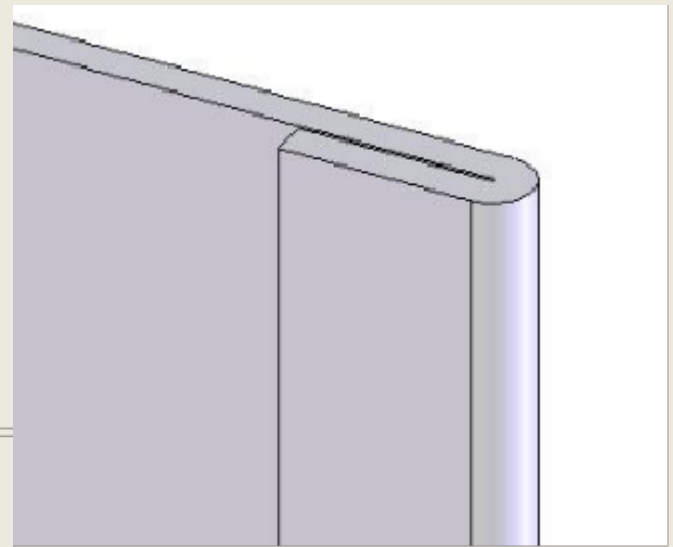


zig-zag pattern



Joining Sheets – Seams

Hemming and seaming are two similar metalworking processes in which a sheet metal edge is rolled over onto itself.



Hand Seams

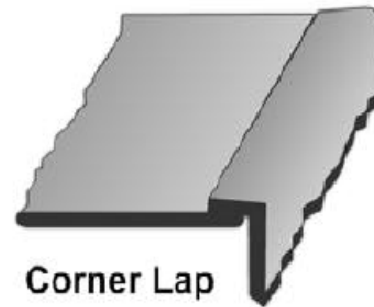
The lap seam is a basic metal seam where one piece overlaps the other. A "lap" is created by making a fold across the edge, then laying the lap over the other piece of metal, pounding flat and soldering, welding or securing with rivets.



Plain Lap



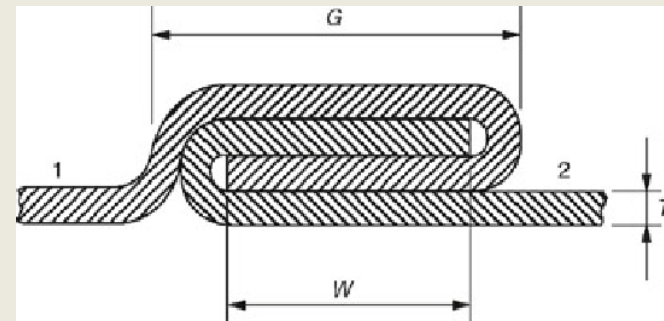
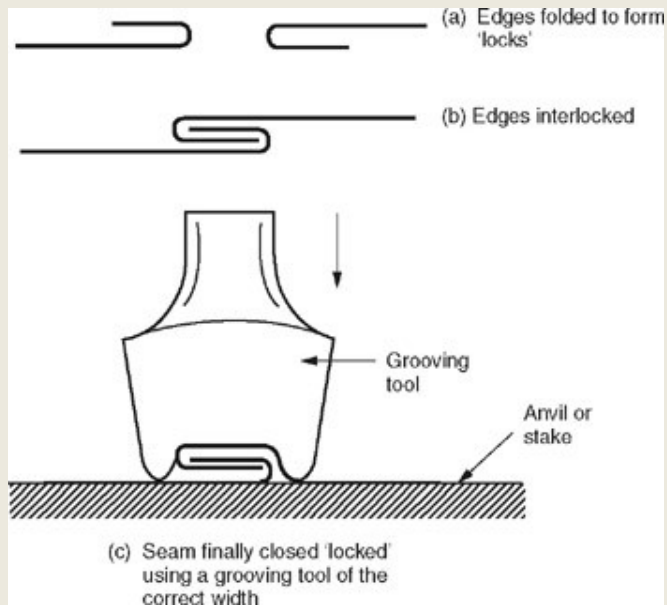
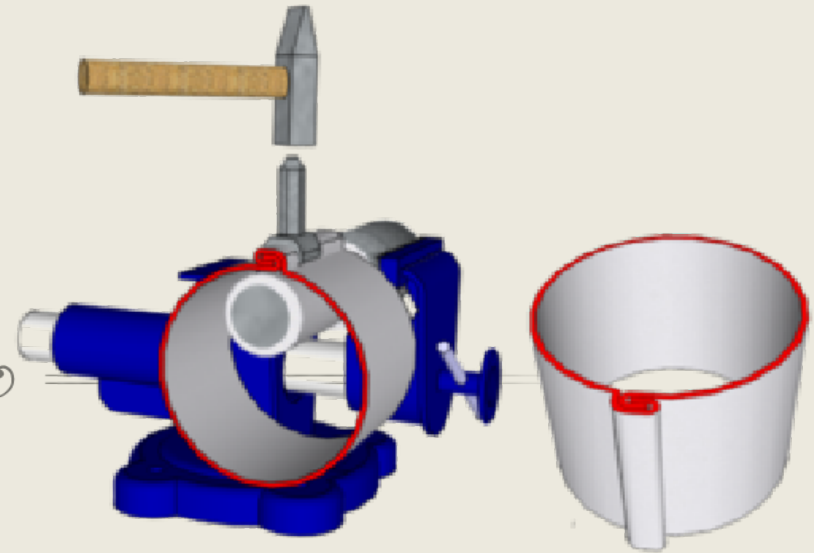
Off-Set Lap



Corner Lap

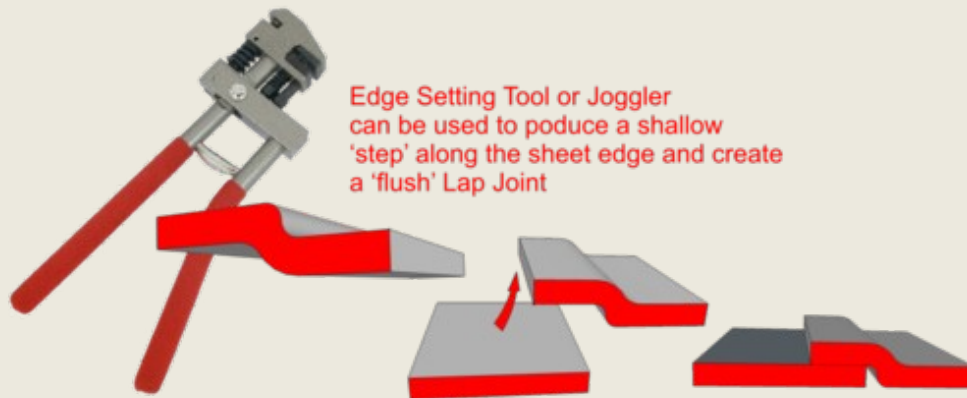
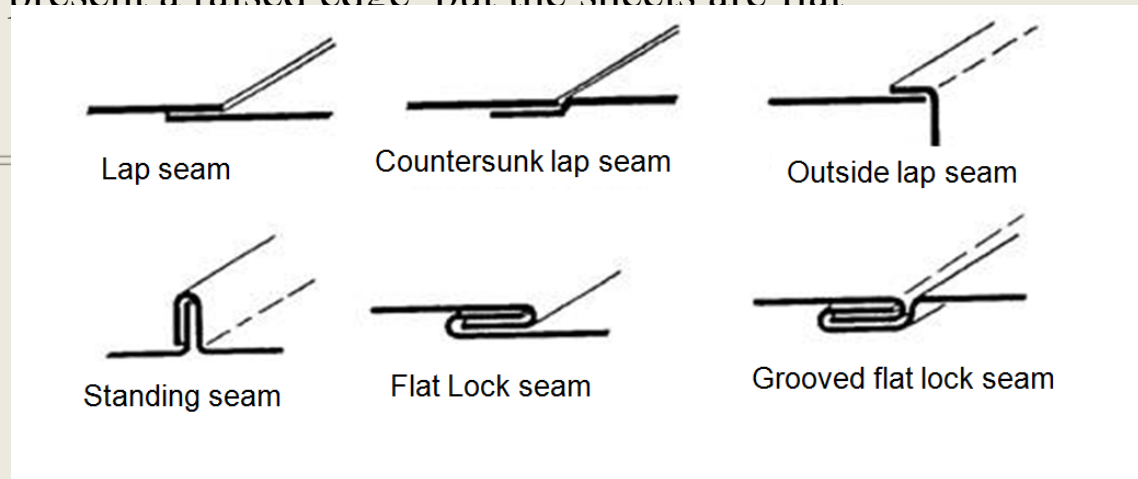
Grooved seam/ Flat Lock Seam

The grooved seam process is similar to the folded seam process; it involves connecting two folded edges, usually made with a machine, which are interlocked and pressure-tightened to make the seam.



Countersunk lap seam

When a lap seam is formed with an edge setting tool or a jogger, a countersunk lap seam is made. It appears nicer and presents a raised edge, but the sheets are flat.



Machine Seams

Machine seams in sheet metal work and hand seams appear the same way, but machine seams are quicker to make and used in production. Many seaming techniques today are accomplished using machine seams.



Lap seam



Countersunk lap seam



Outside lap seam



Standing seam



Flat Lock seam



Grooved flat lock seam

Soft Soldering

Clean the parts

Apply the flux

Preheat solder iron

Dip in flux

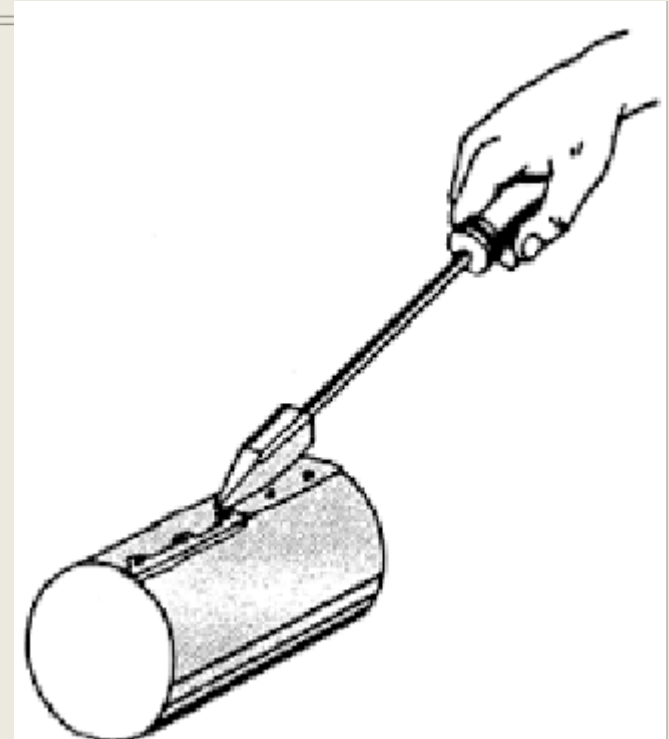
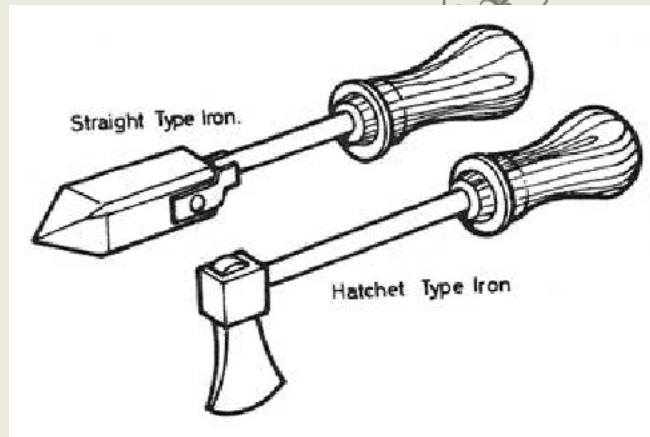
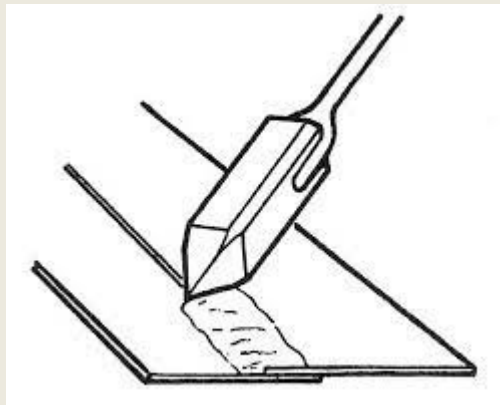
Solder

Reheat

Repeat

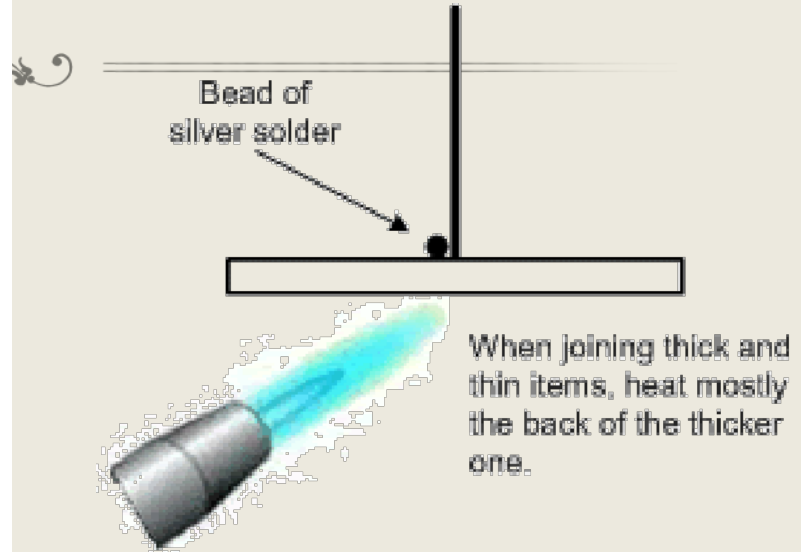
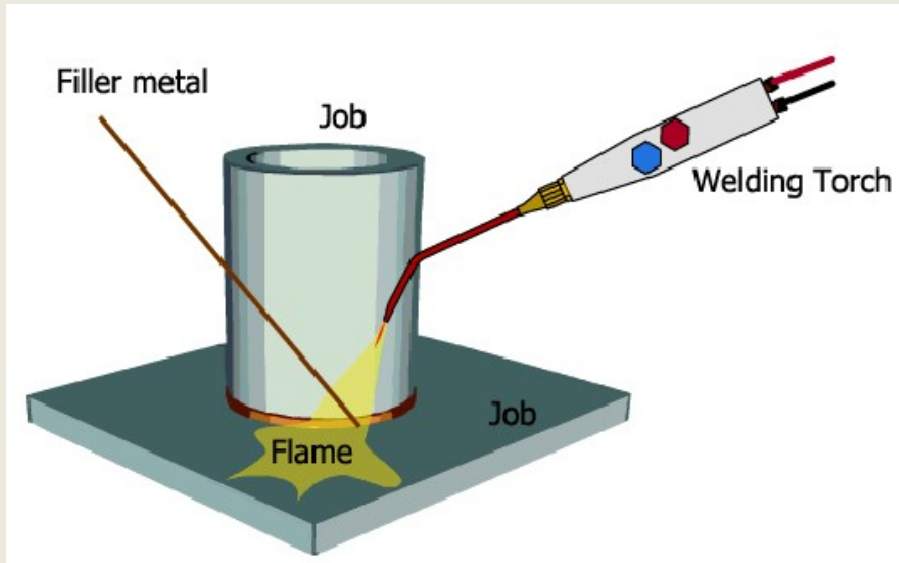
Let it cool

Clean



Silver Soldering

Silver solder is silver-based filler metal that has been used for years by welders



VIDEOS FOR MATERIALS



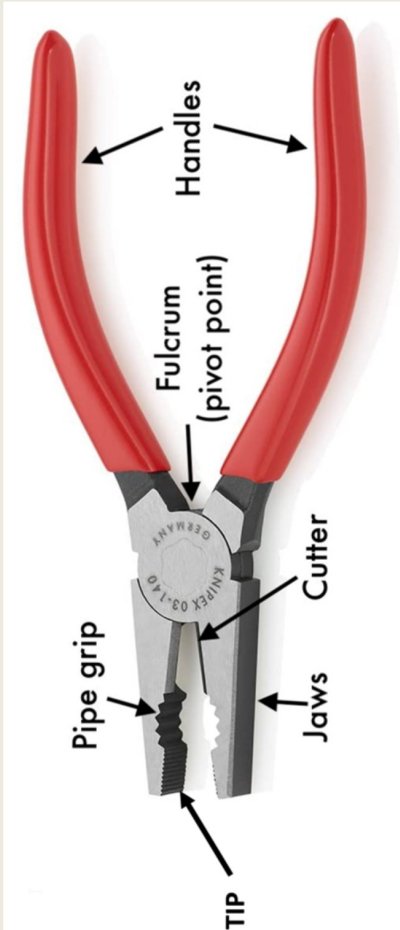
Non Ferrous metals_Types_Properties_Applications_Engineering materials_Aluminium_Zinc_Copper_GTU.mp4







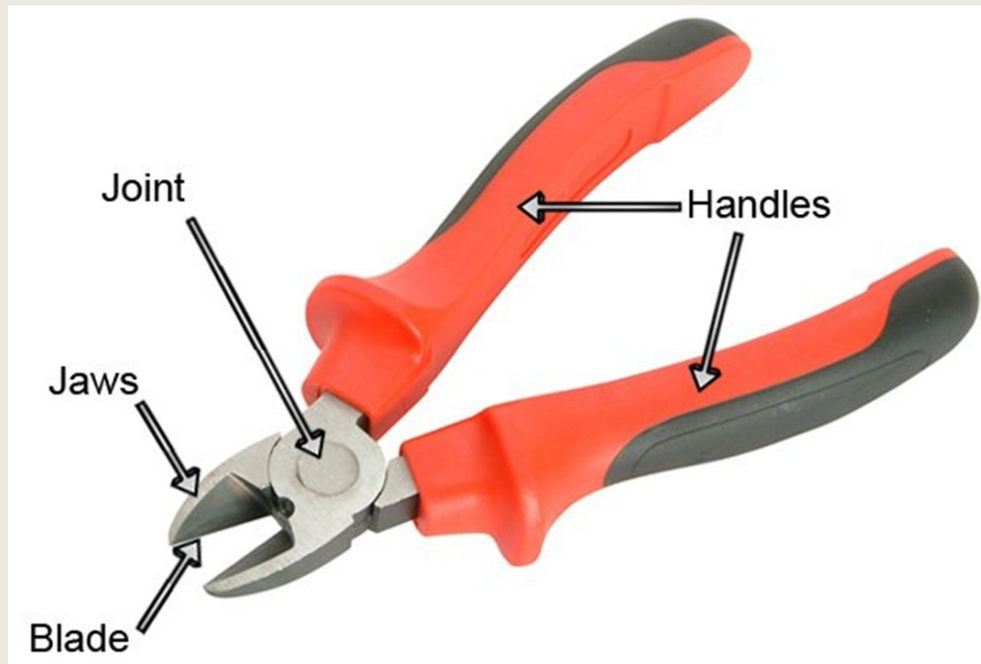
General Pliers



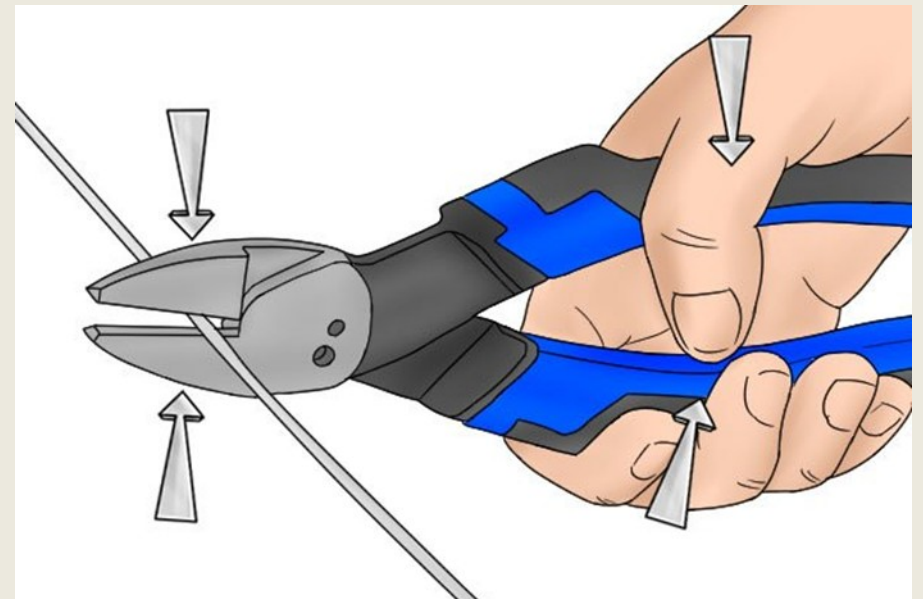
- Tip: The tip is used to pinch or hold an object with precision.
- Pipe Grip: The rounded shape of the pipe grip allows you to hold on tight to rounded objects, such as pipes.
- Jaws: The jaws has a specific size and determines the size of the pliers.
- Fulcrum: This is the pivot point around which the parts of the pliers rotate.

Some pliers also have a grab or pinching point just behind the fulcrum.

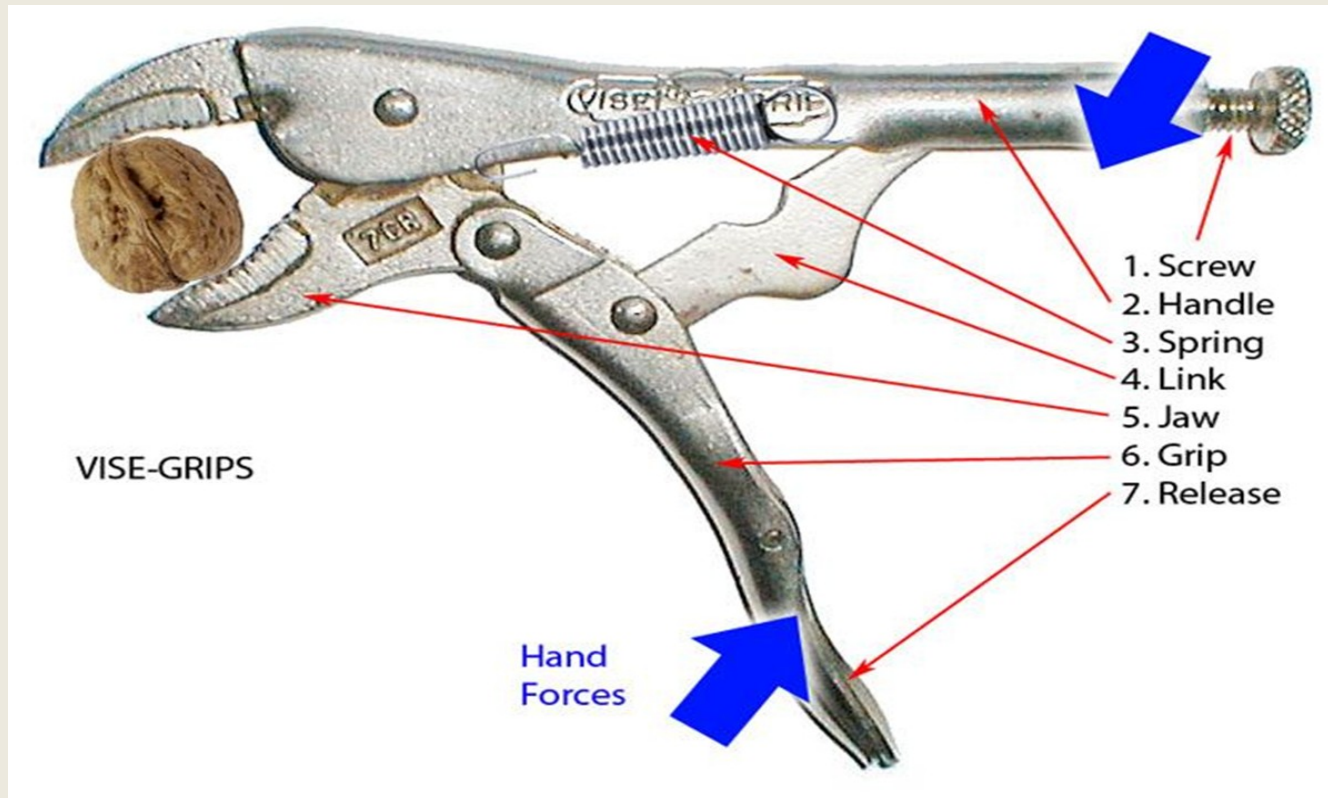
- Handles: These are used to hold onto the pliers. Some pliers have insulated handles so the pliers can be used on electrical wiring.

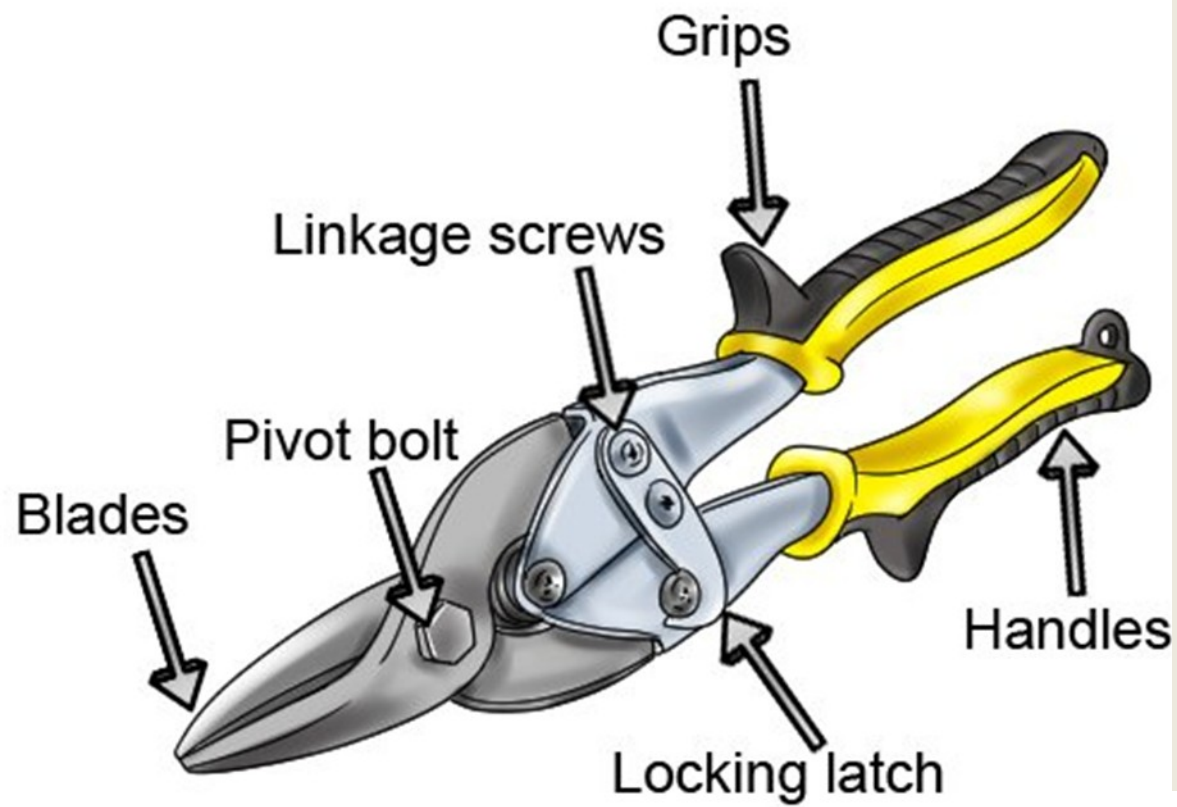


Side Cutter



VICE GRIP

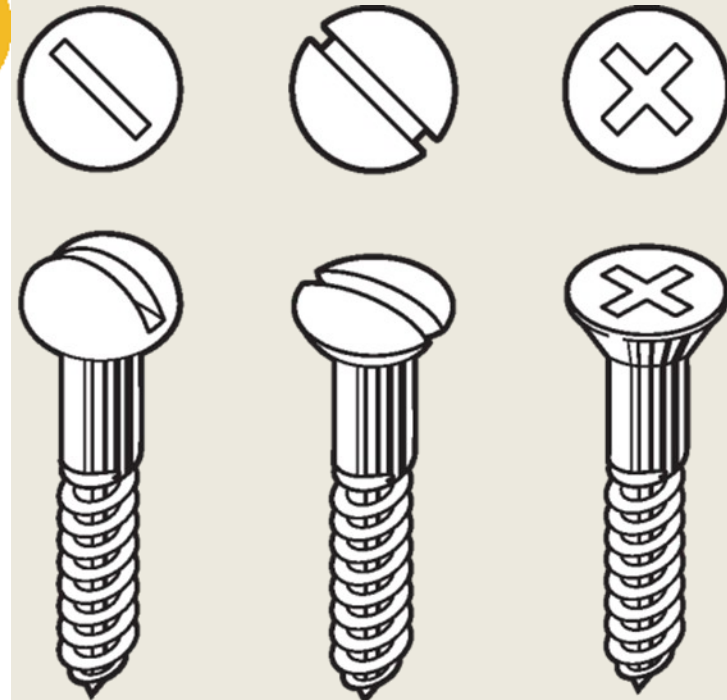
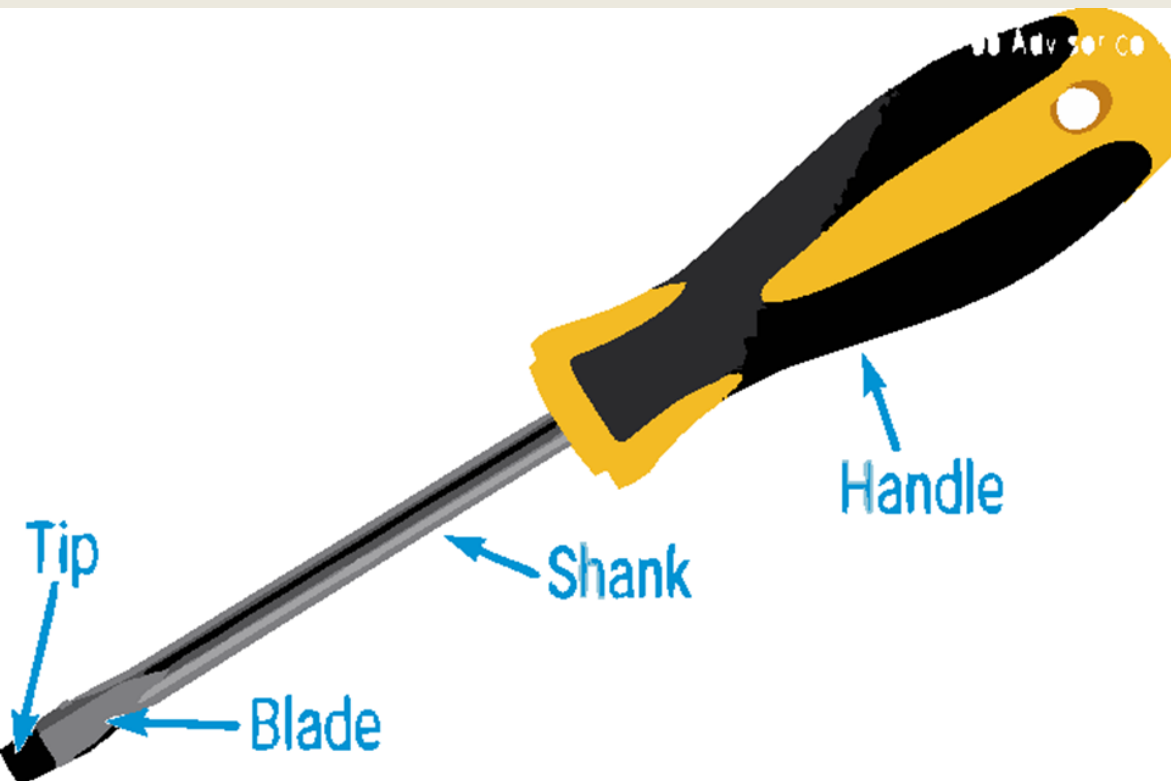




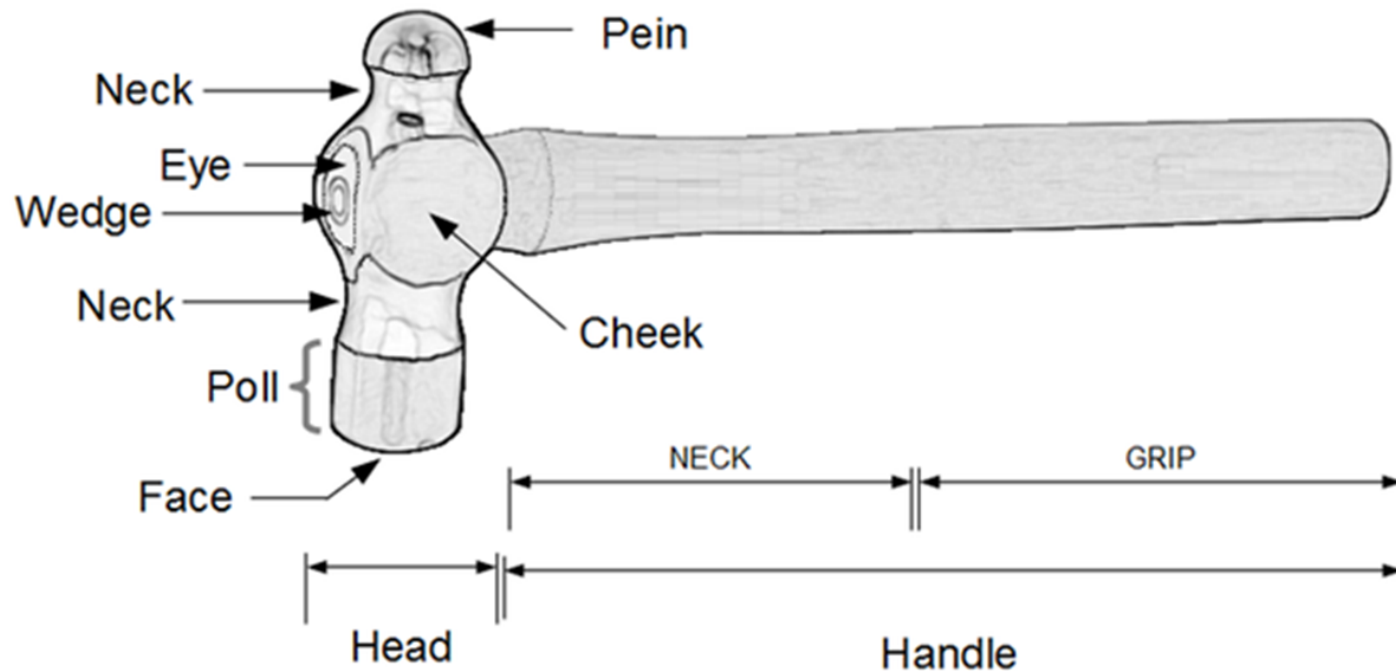
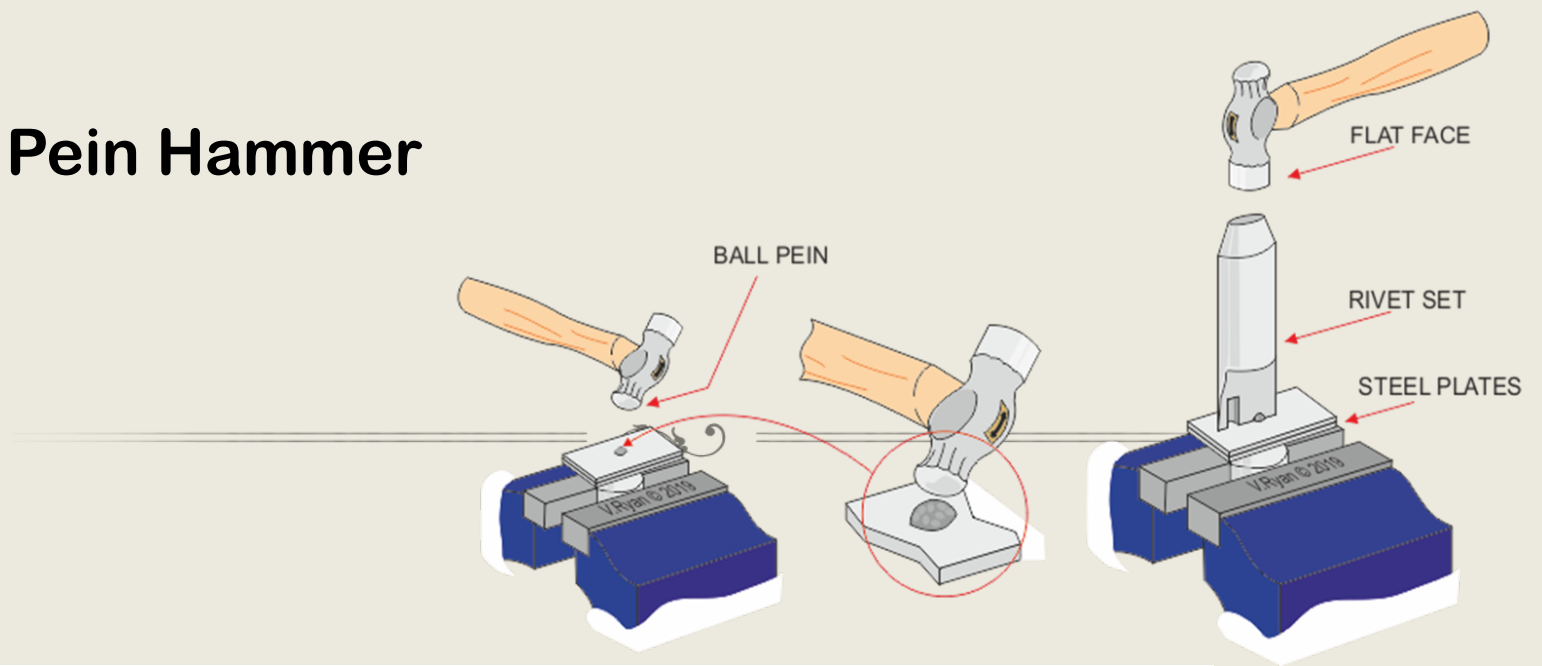
Tin snips



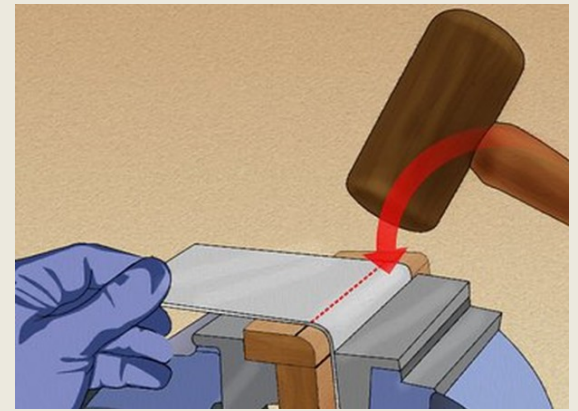
SCREW DRIVERS



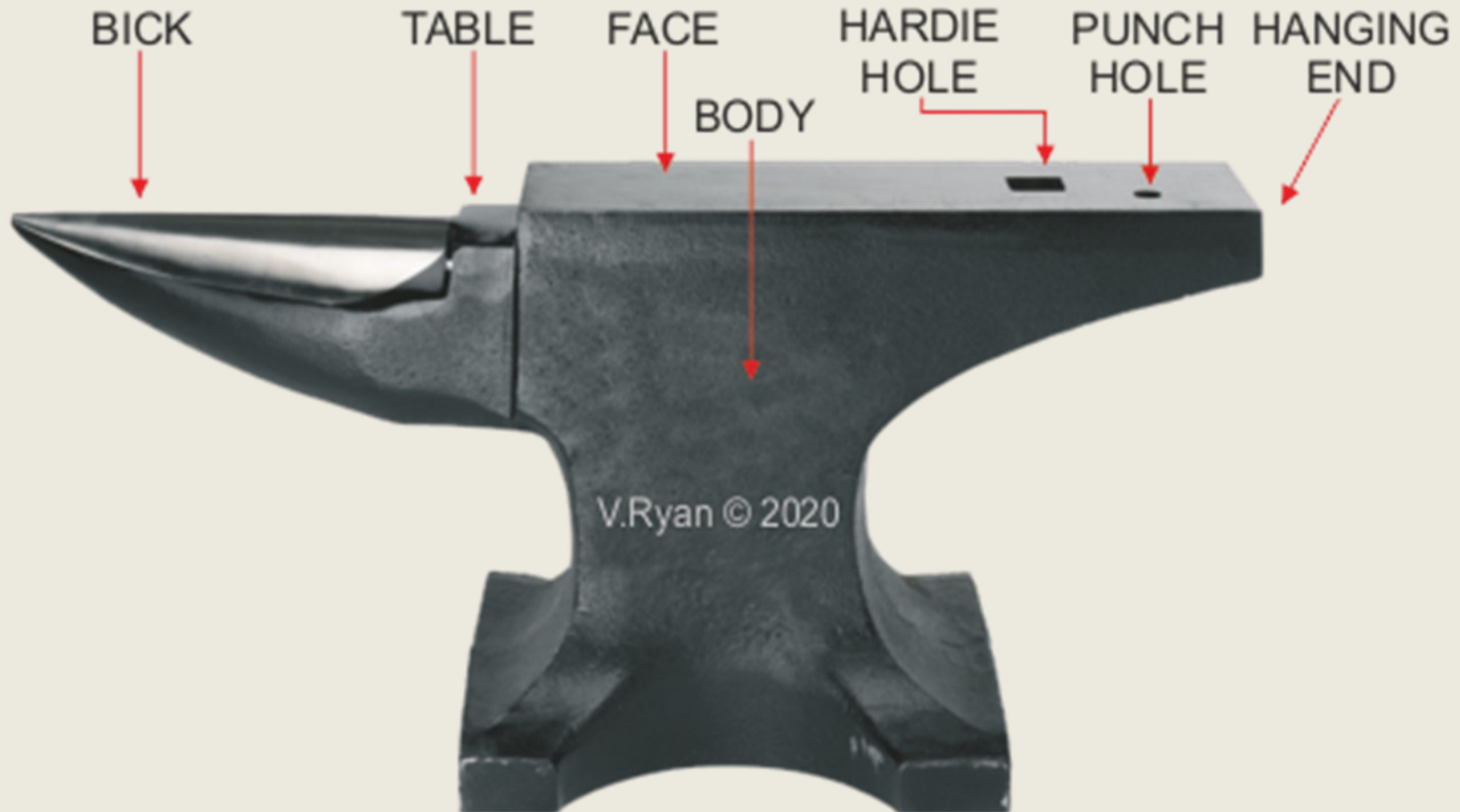
Ball Pein Hammer



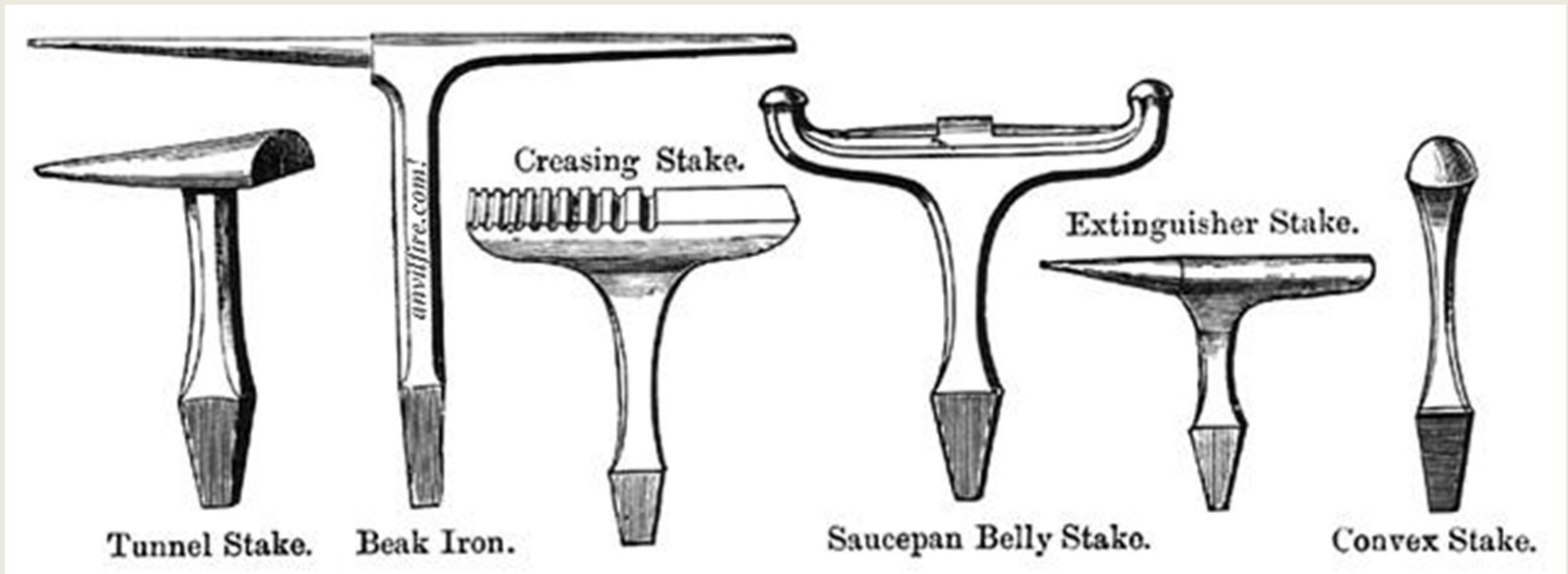
MALLET



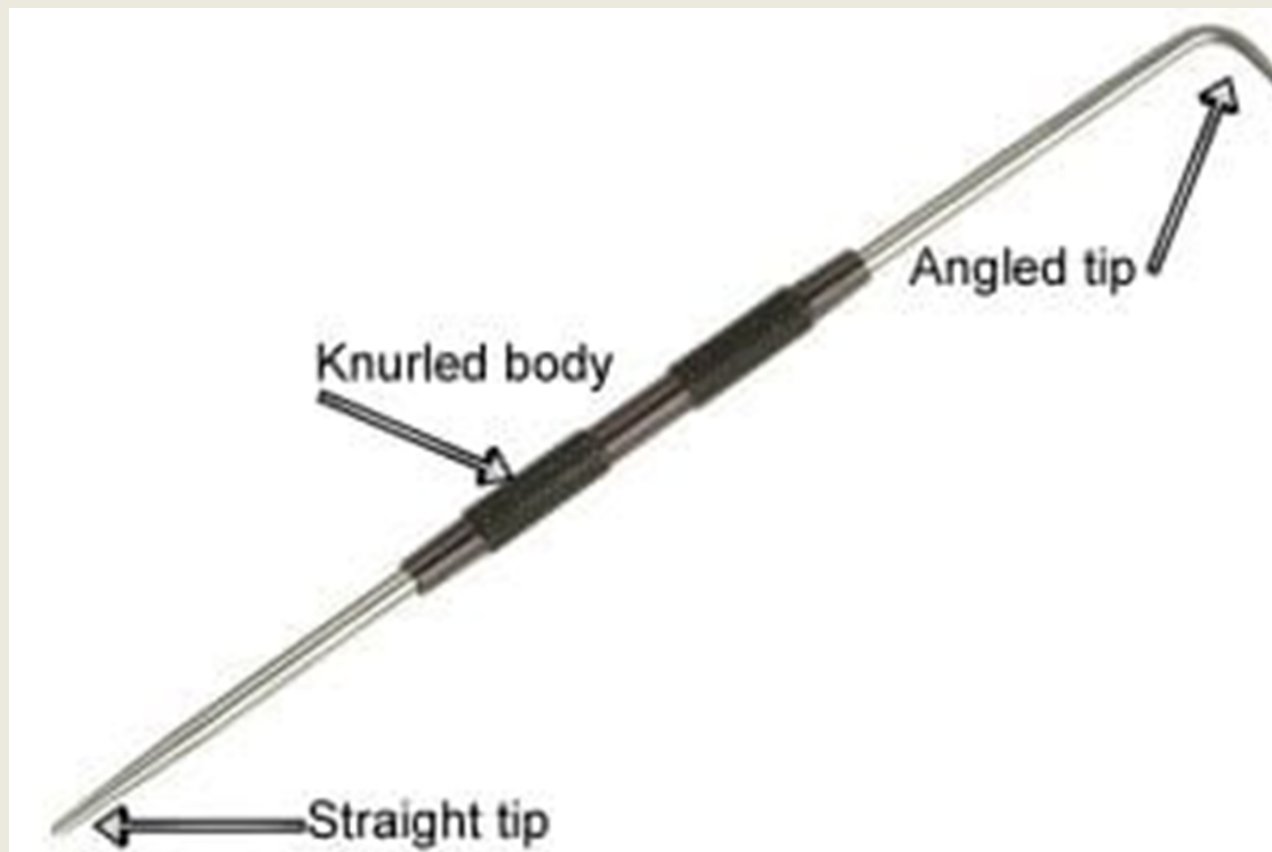
THE ANVIL



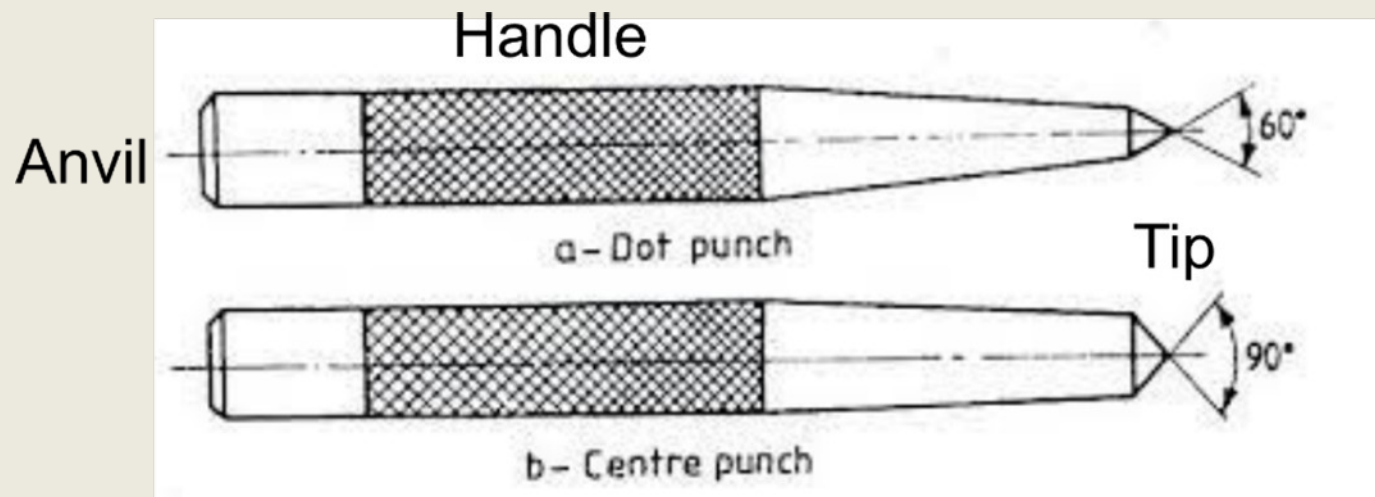
STAKES

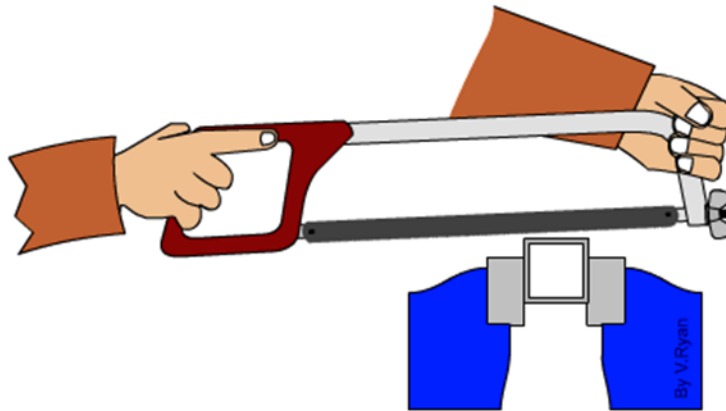
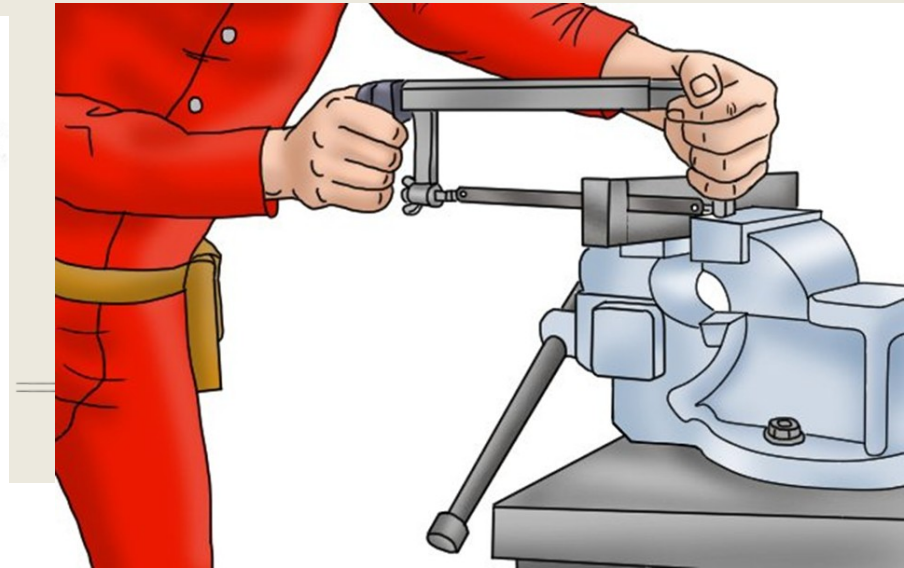
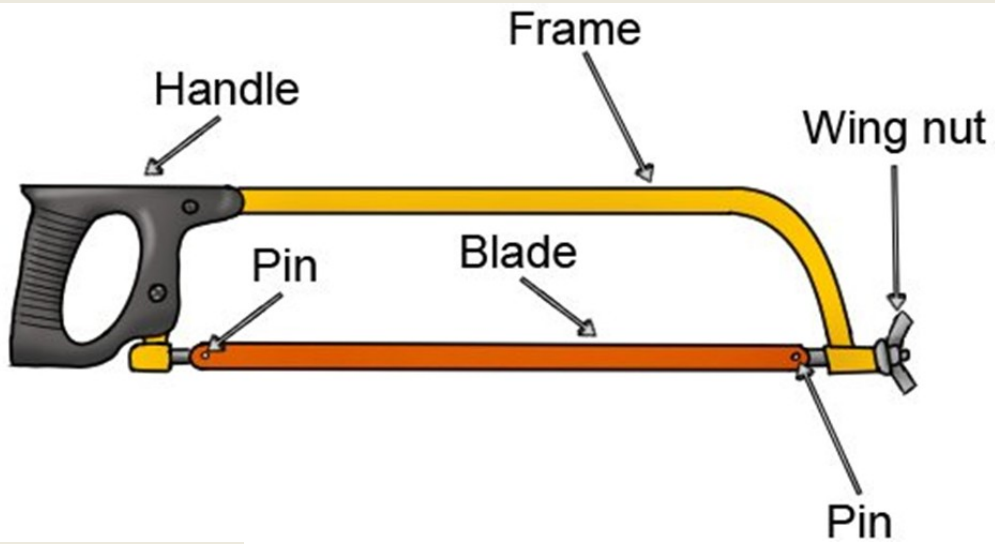


THE SCRIBE

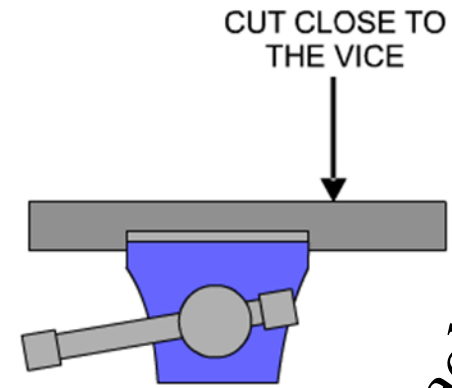


PUNCH





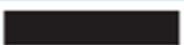
SIDE VIEW

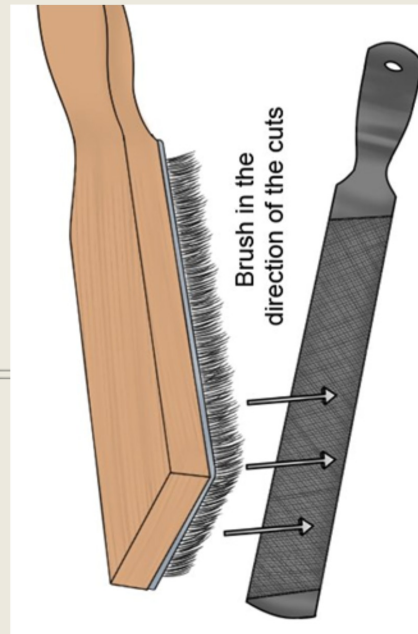


FRONT VIEW

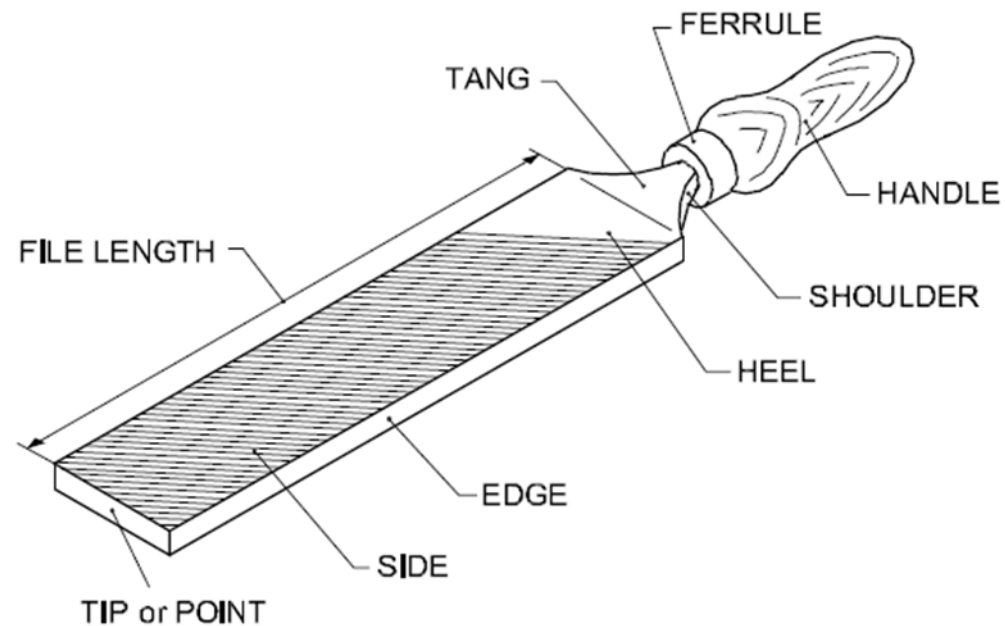
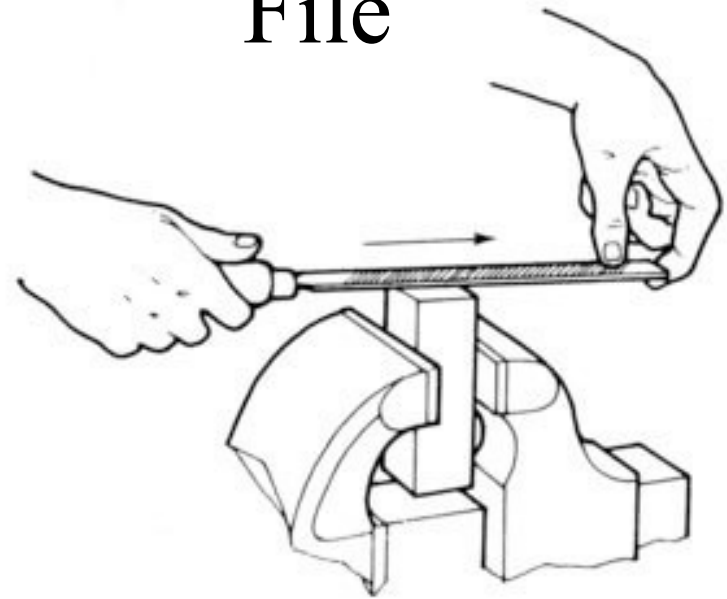
Hacksaw

FILE PROFILES

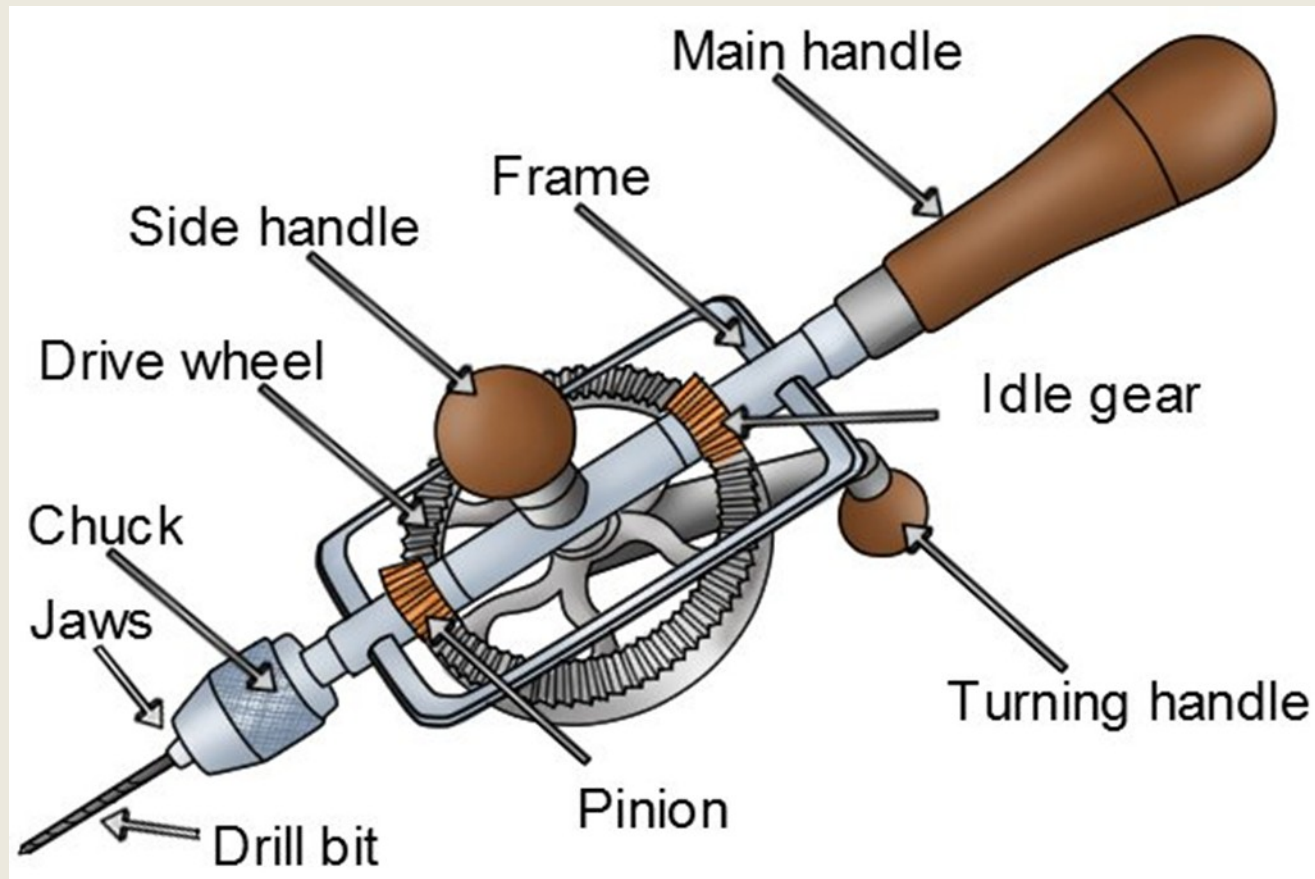
Profile	Use
	Smoothing large curves or flat areas
	Roughing out or shaping large inside curves
	Tight areas or hard to reach spots
	Shaping or enlarging holes, or shaping tight inside curves
	Shaping inside corners
	Smoothing inside curves
	Tight areas where clearance is a problem



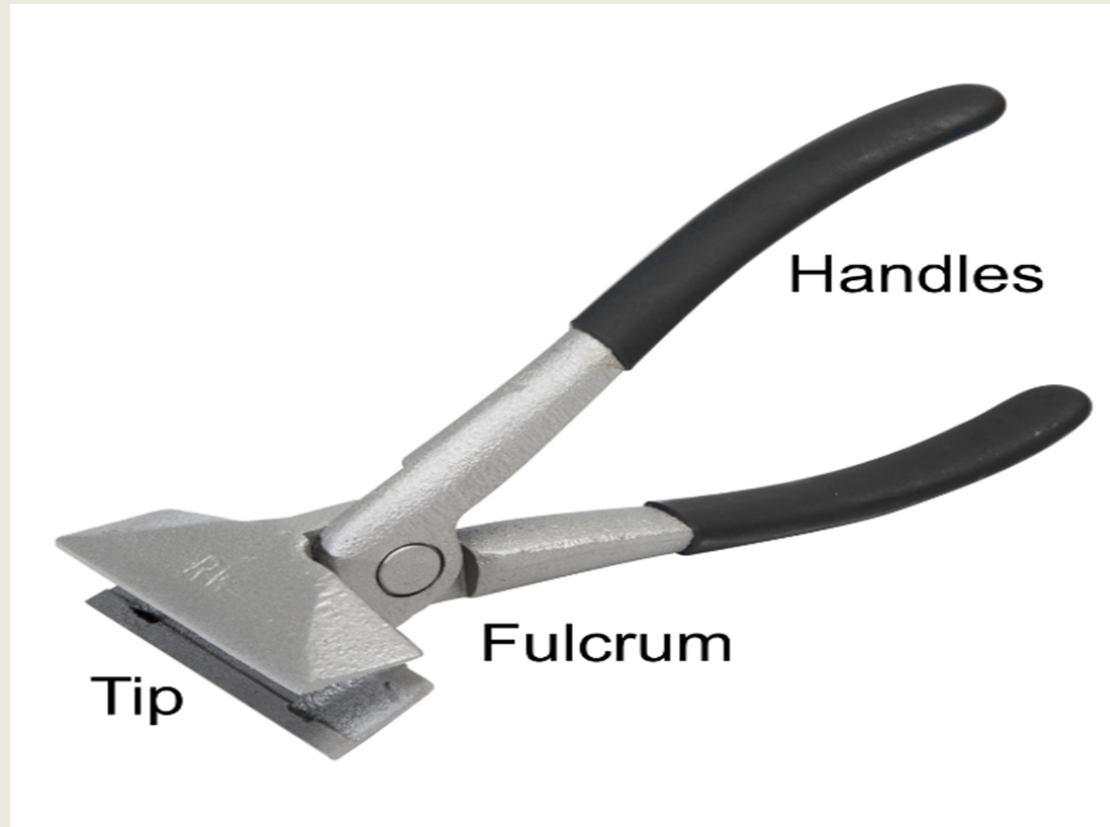
File



HAND DRILL



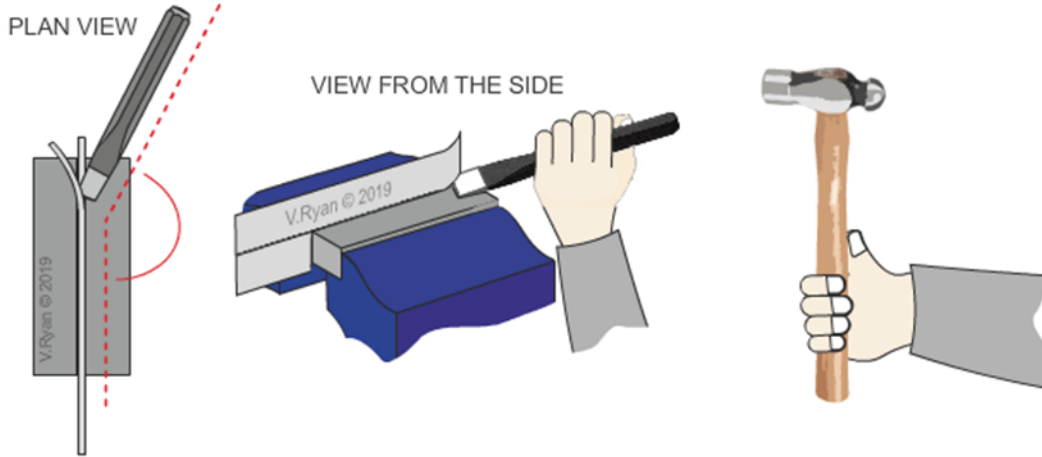
HAND SEAMER



Cold Chisel

PLAN VIEW

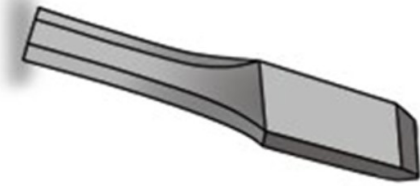
VIEW FROM THE SIDE



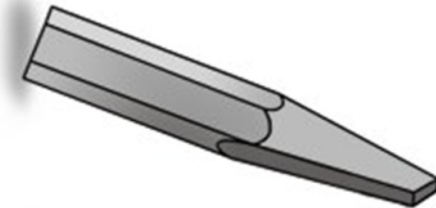
Diamond
Point
chisel



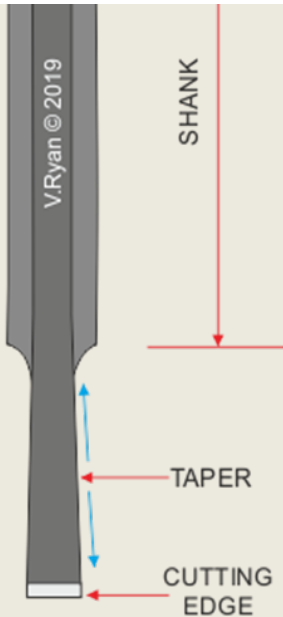
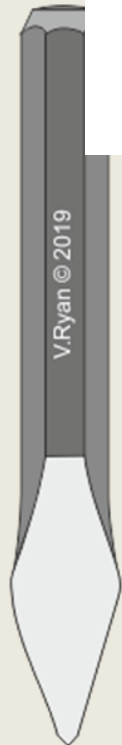
Flat
chisel



Half
round
chisel



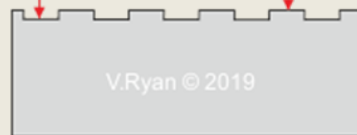
Cross cut/
cape
chisel



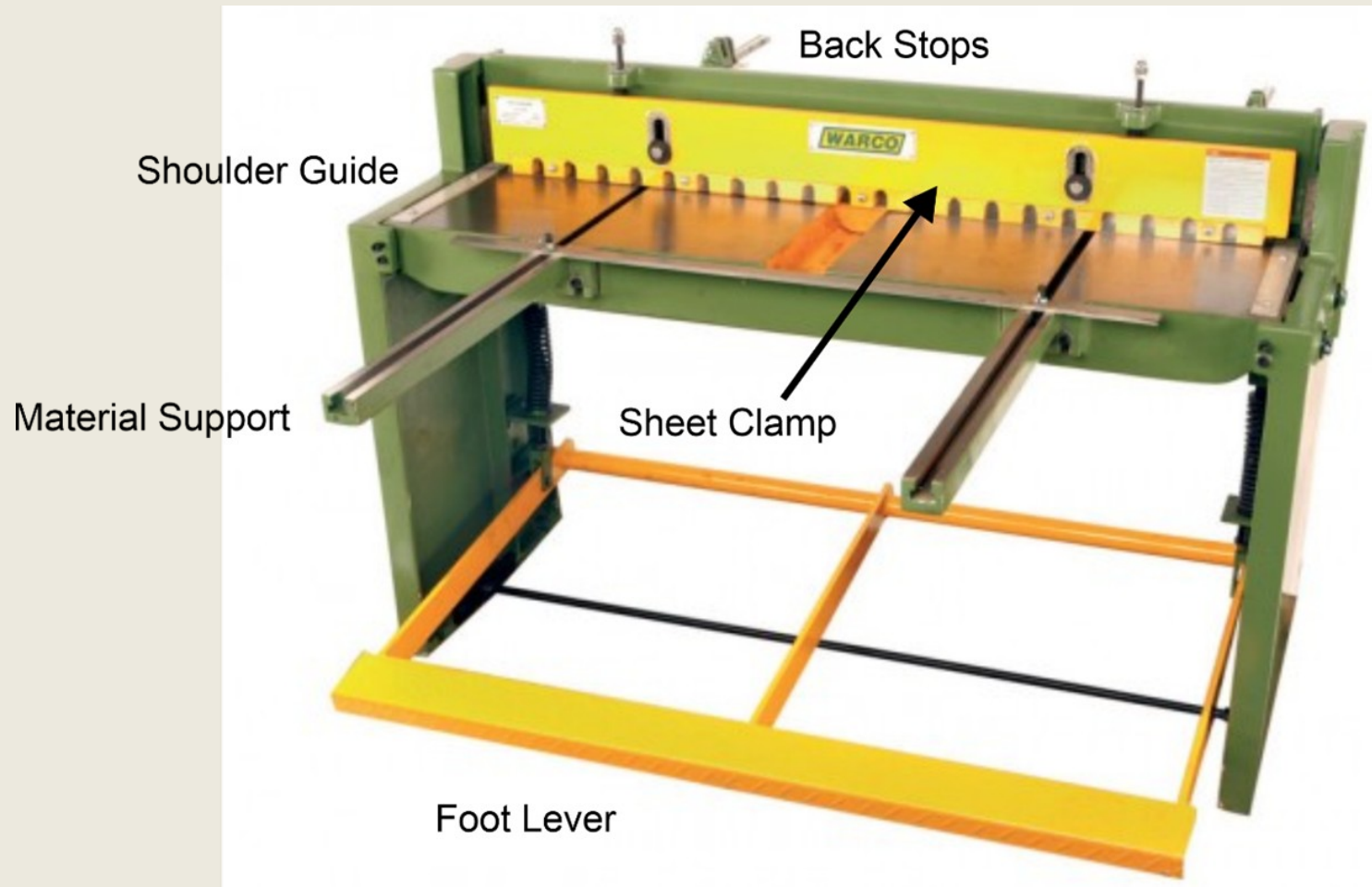
CROSS CUT
CHISEL



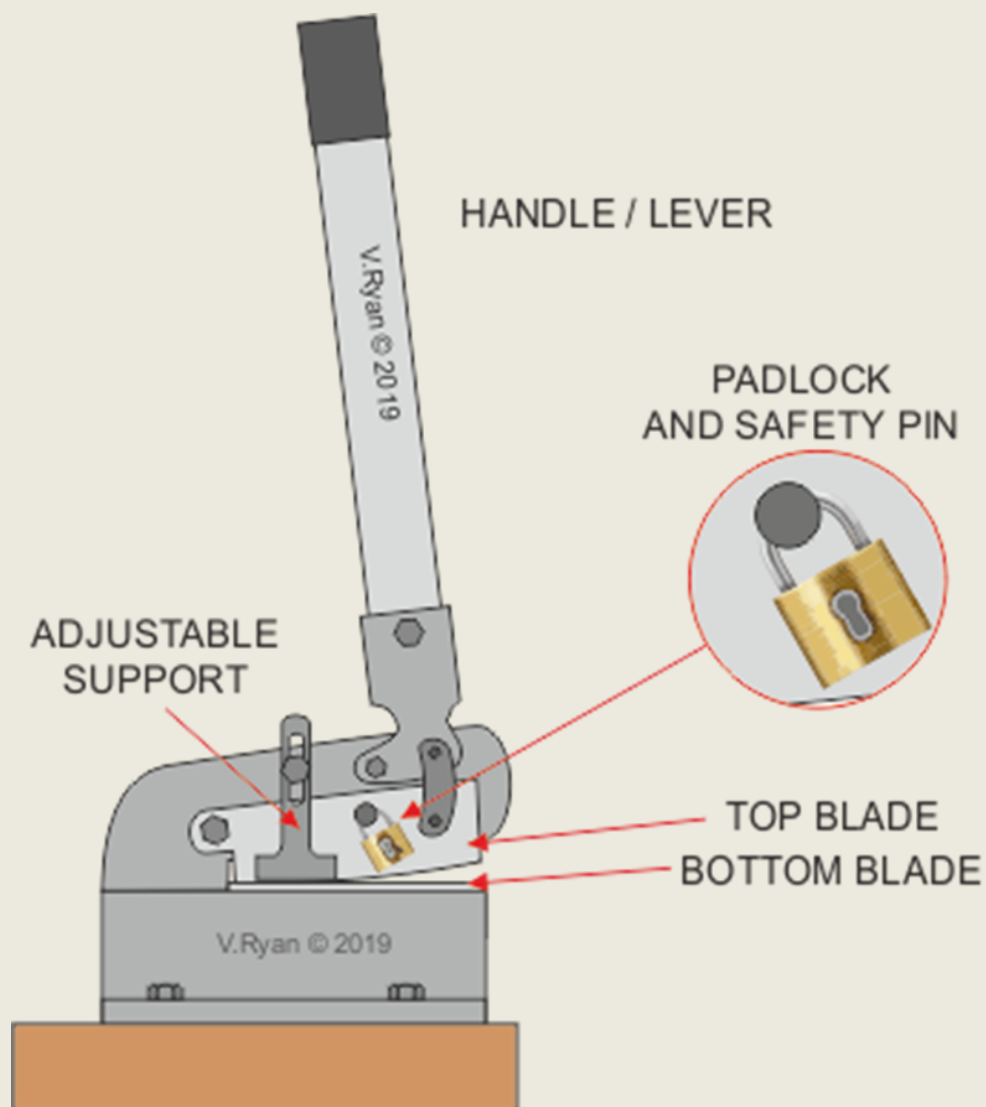
CHISEL



Guillotine: Foot Operated



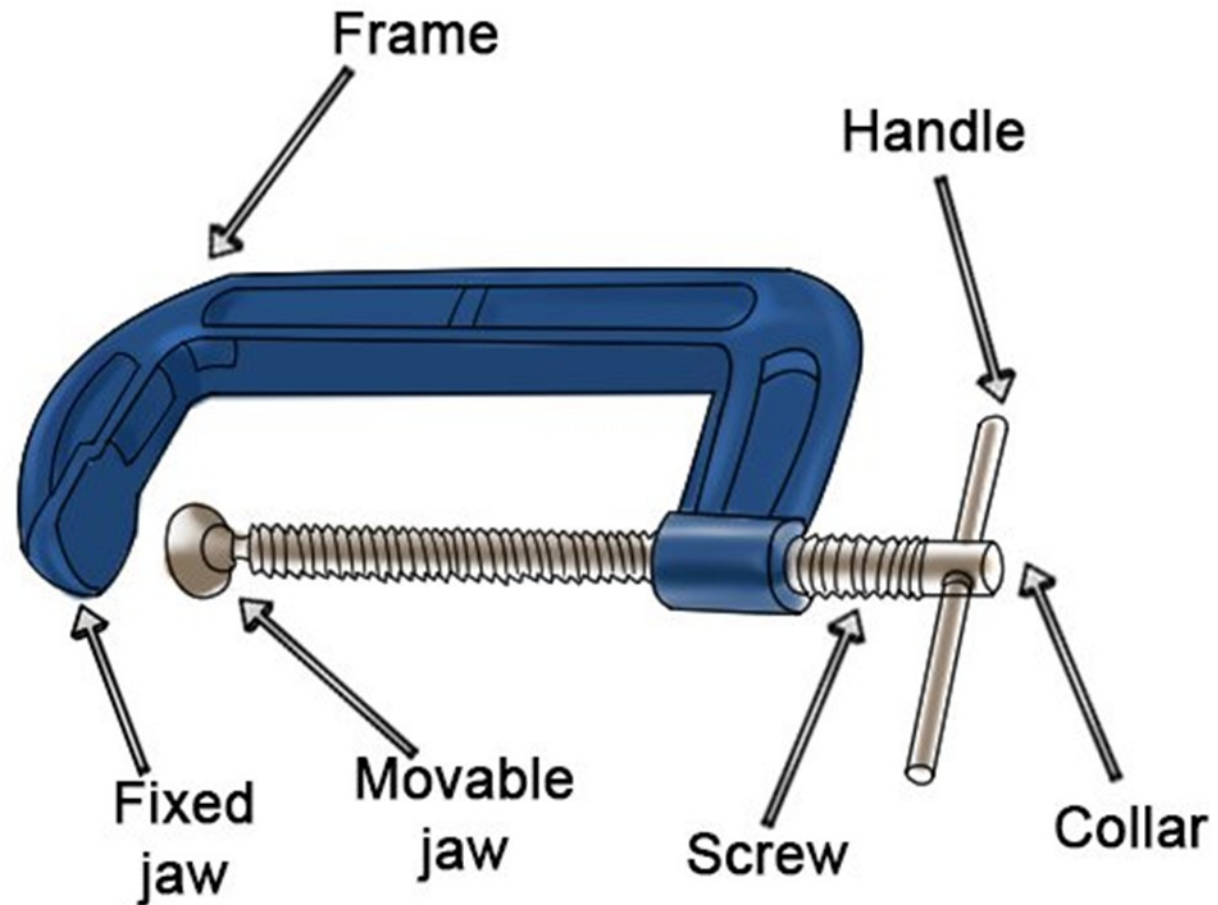
Hand Guillotine / Shear



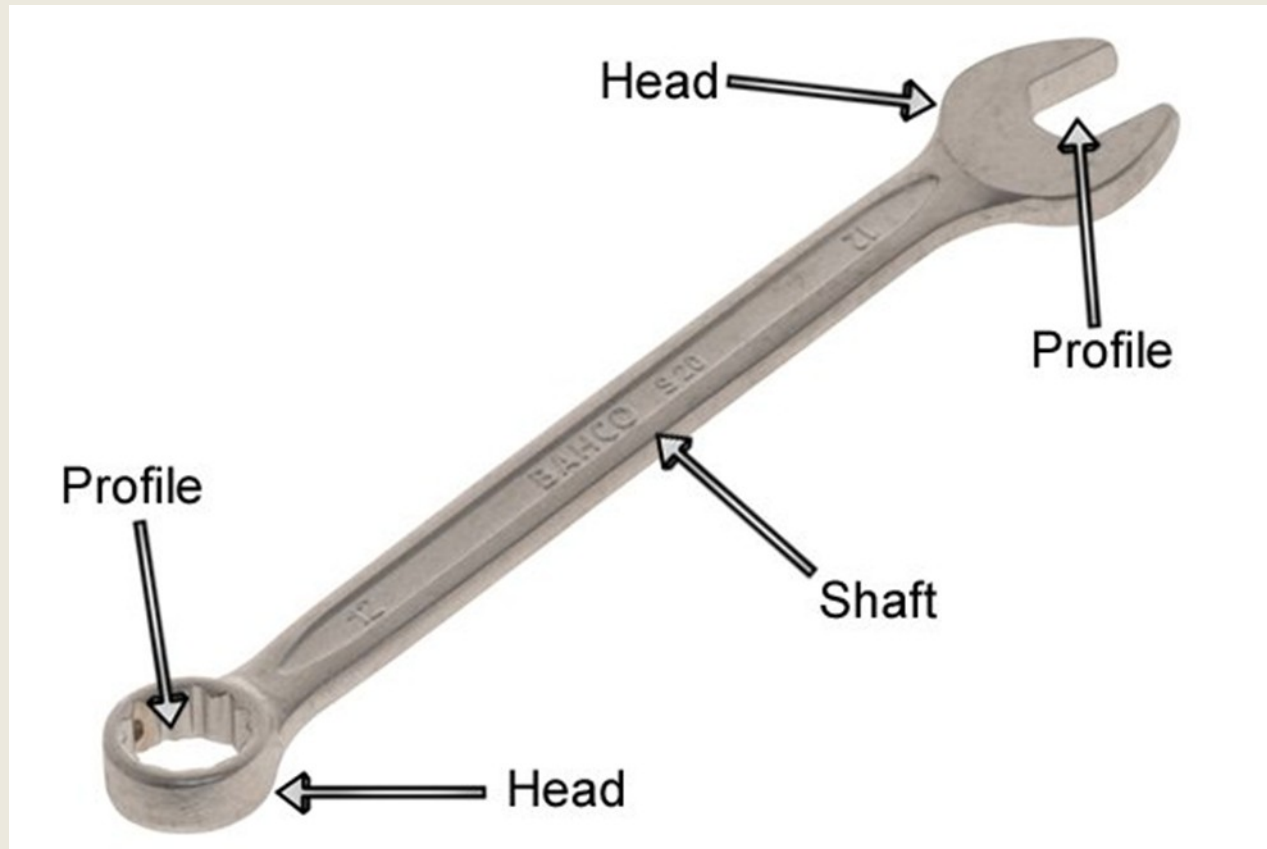
Nibbler



G Clamps



JOINING TOOLS: SPANNERS



JIGS

Jigs are used when a worker needs to manufacture many parts that are the same.

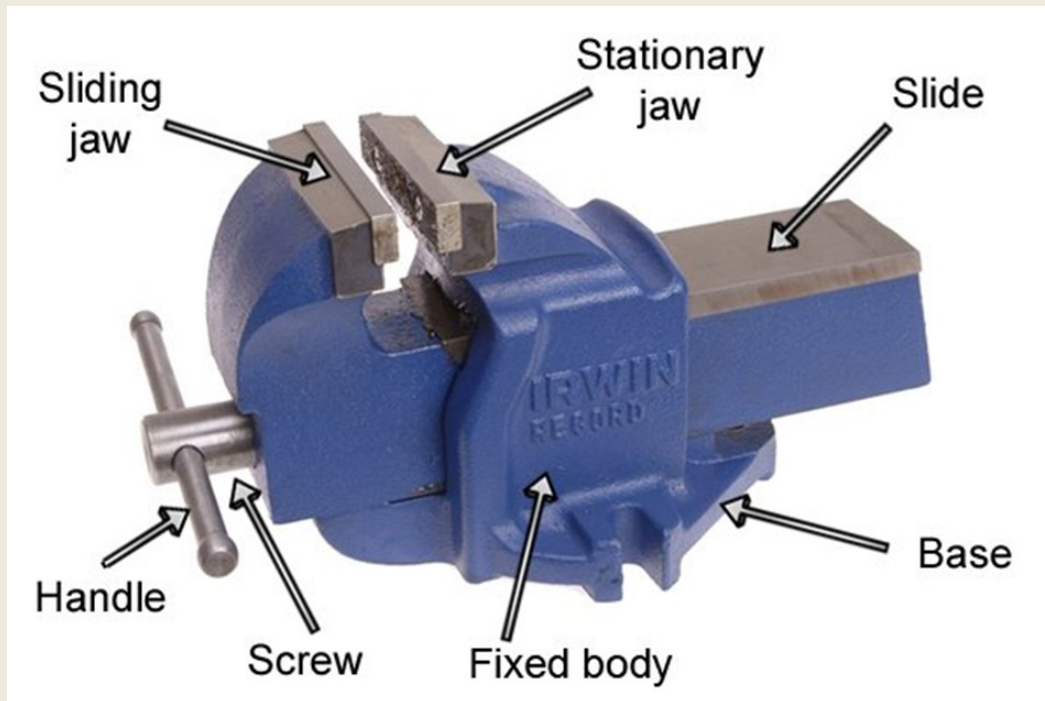
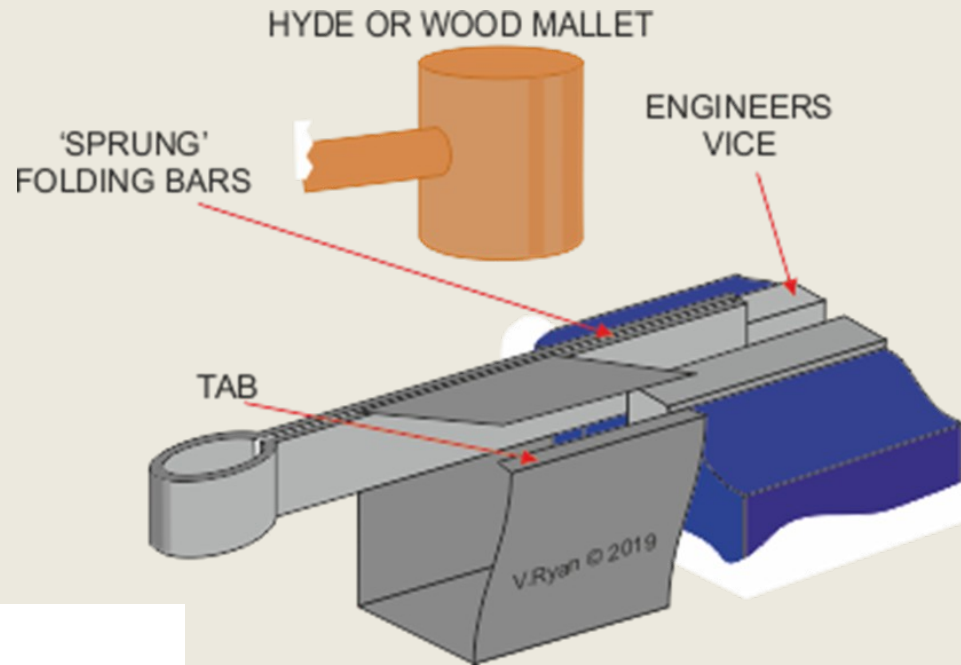
When making similar parts, you will do the same measurements and materials handling time after time.

One way to cut down on manufacturing time and ensuring uniformity in manufacturing is utilising a jig.

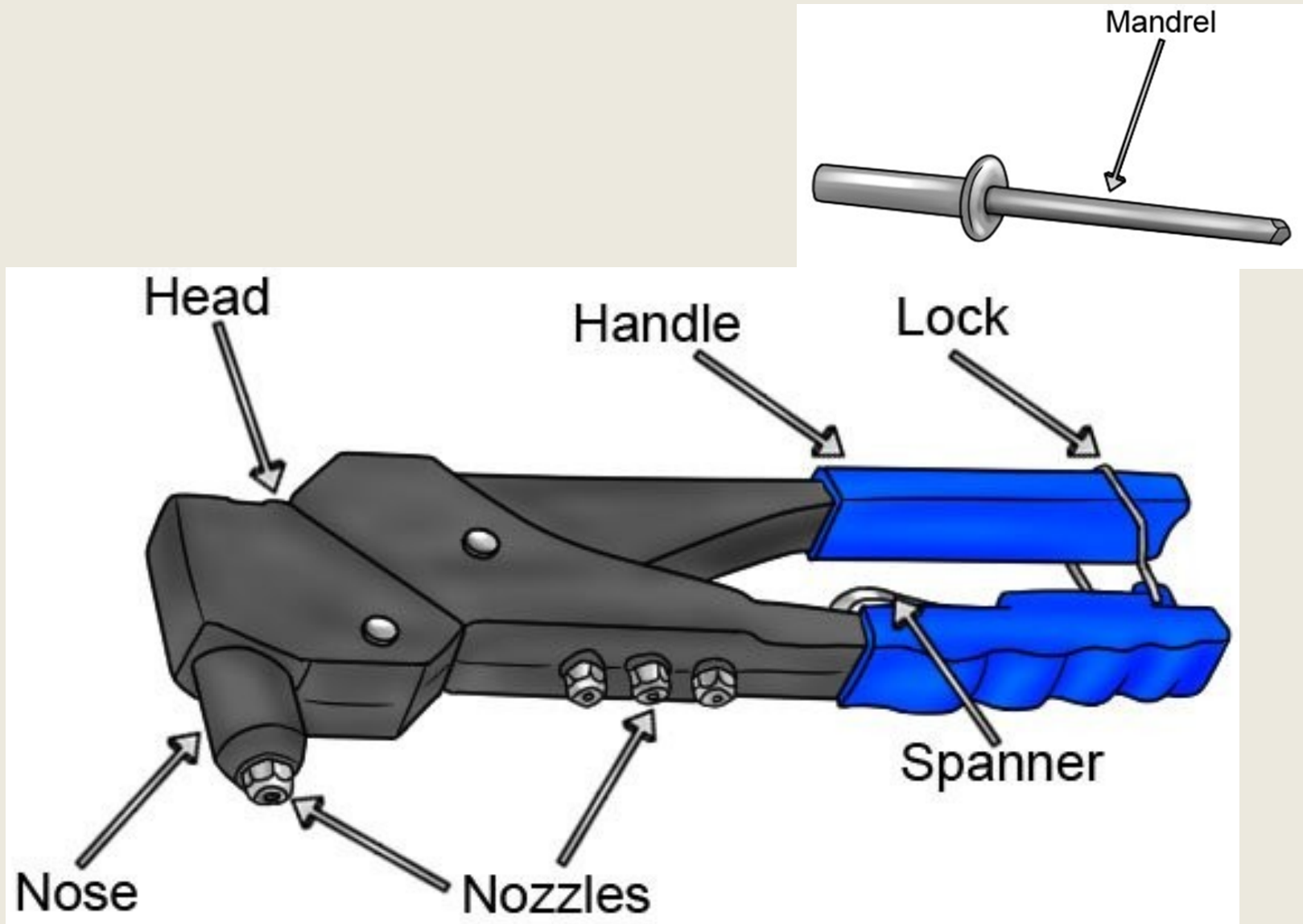
A jig is a pre-formed clamp or positioning points that ensure that the workpiece can be place in the same position time after time, making the same cut or bend time after time



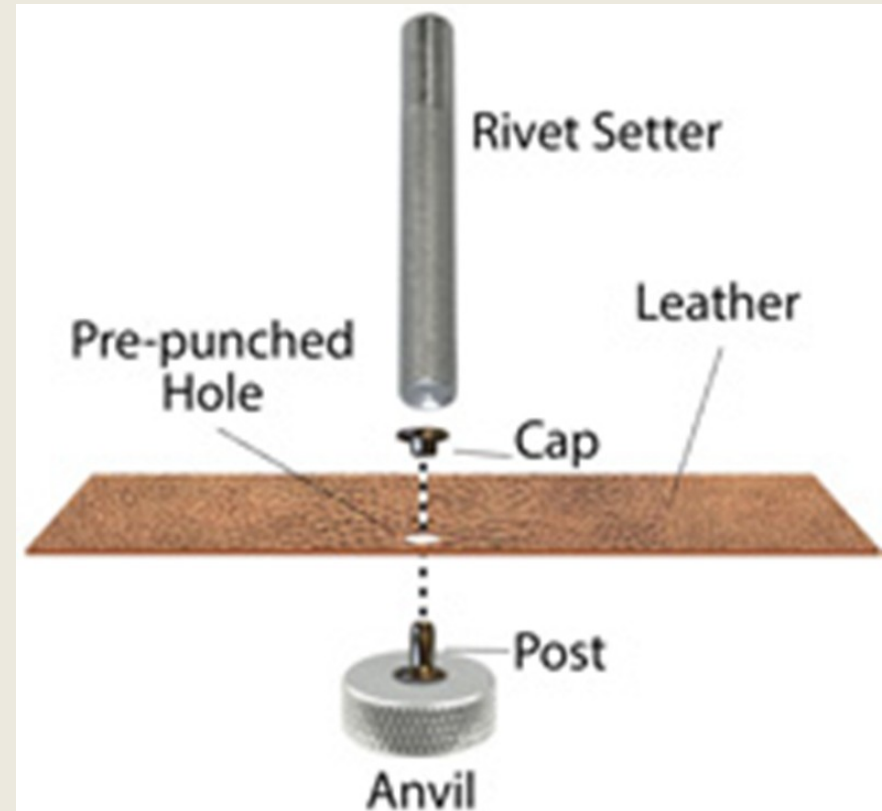
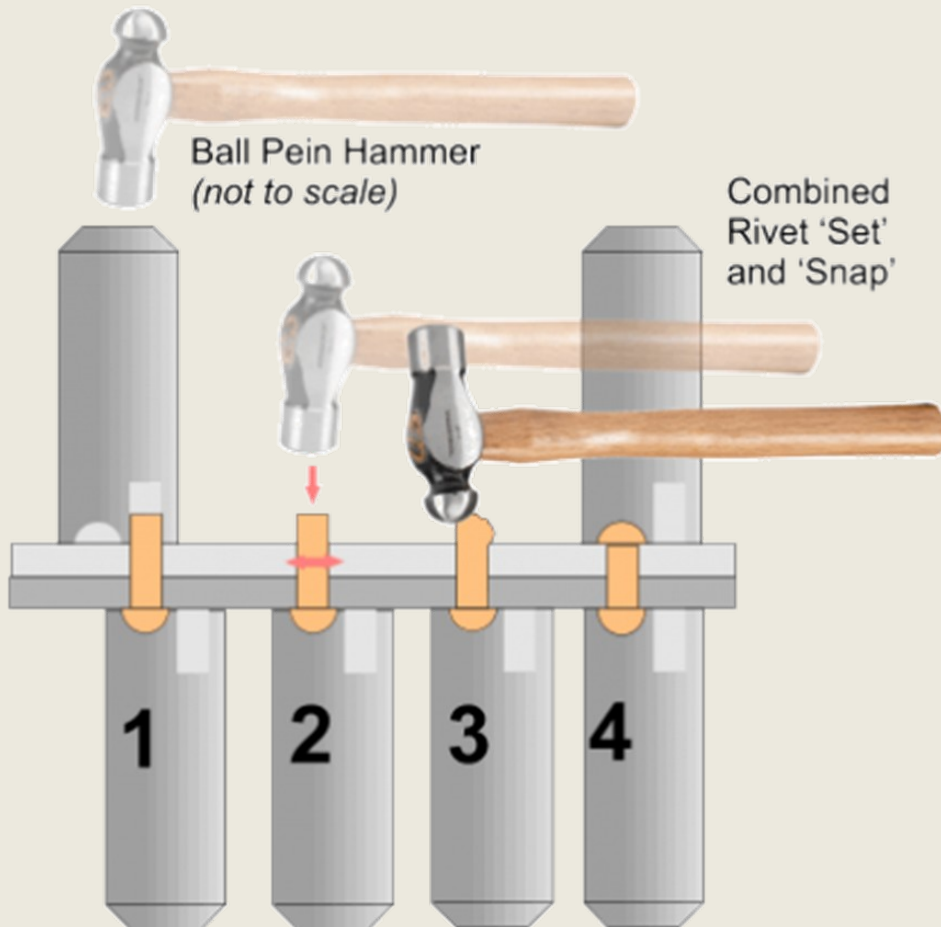
Engineers Vice & Folding Bars



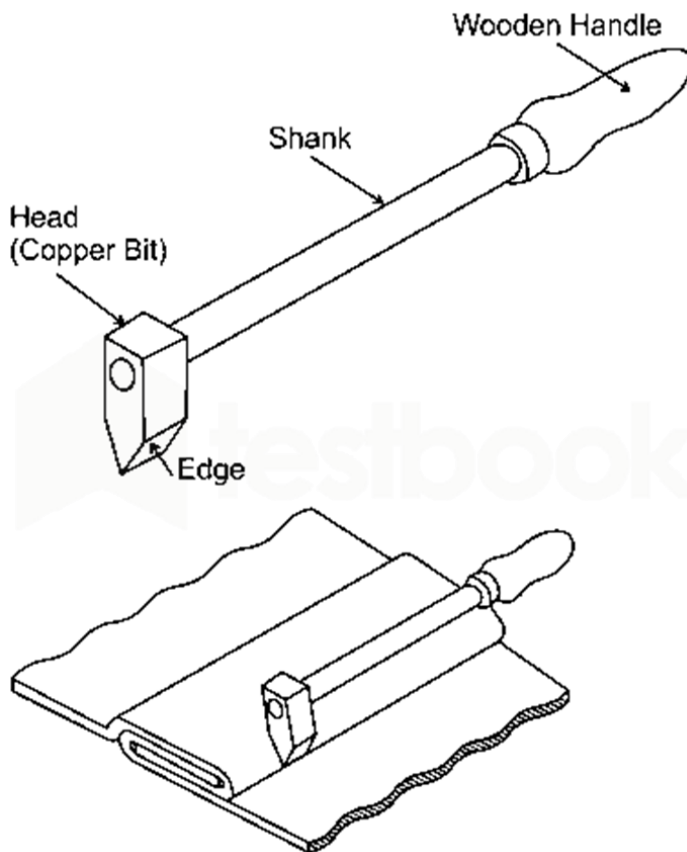
Joining Tools: Rivet gun



Joining Tools: Rivet Snap Set



Joining Tools: Solder Iron



(a) Flanged T



(b) Flush lap



(c) Flanged corner



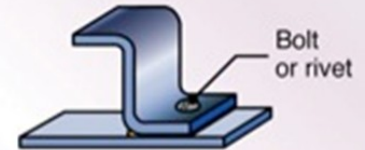
(d) Line contact



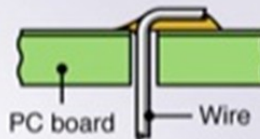
(e) Flat lock seam



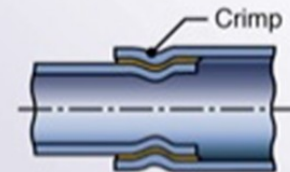
(f) Flanged bottom



(g) Gull wing



(h) Through hole

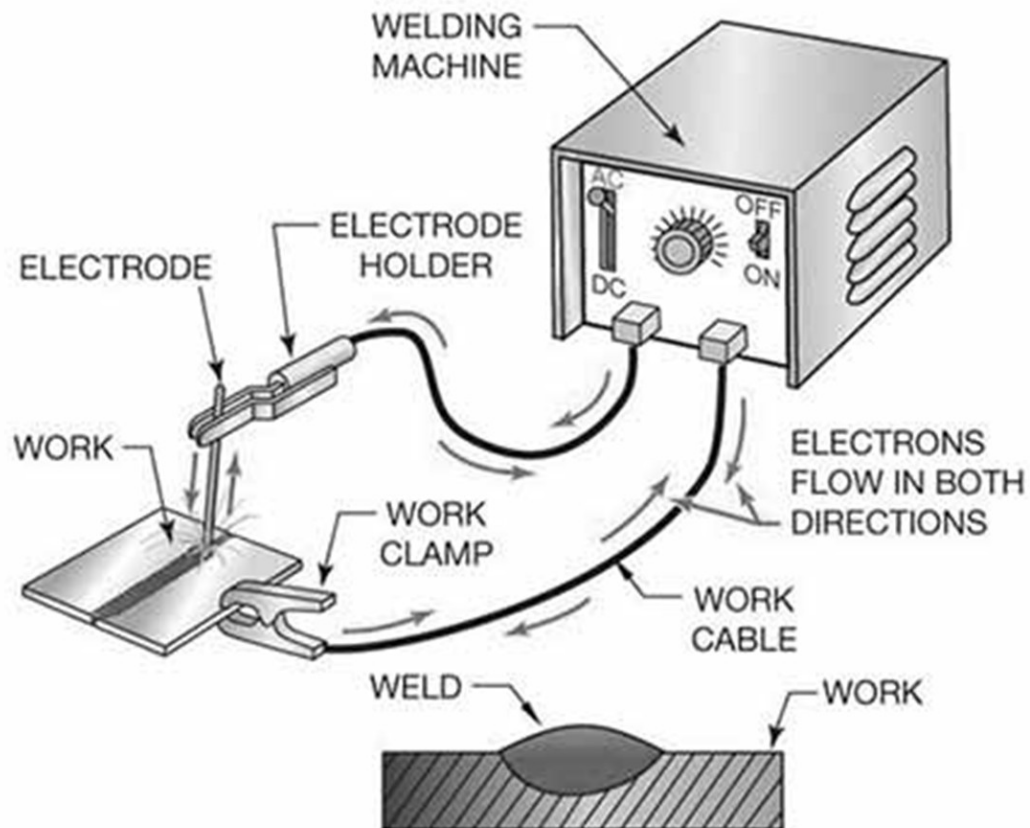


(i) Crimped

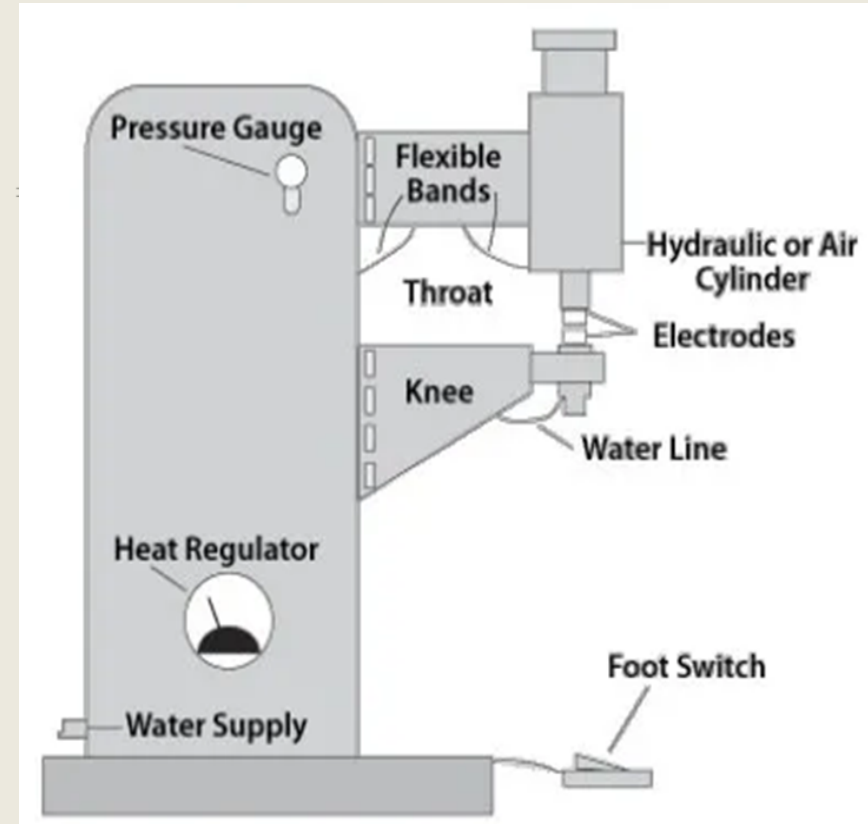
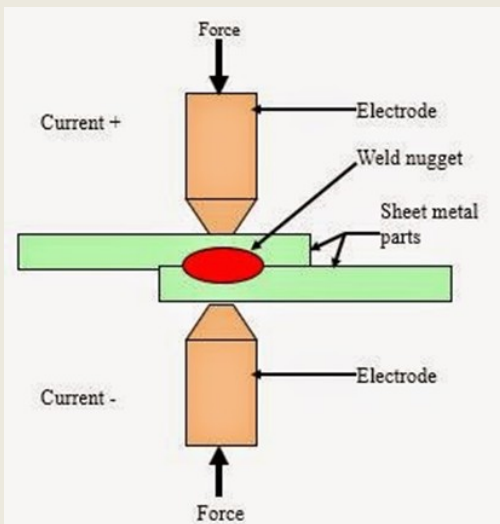
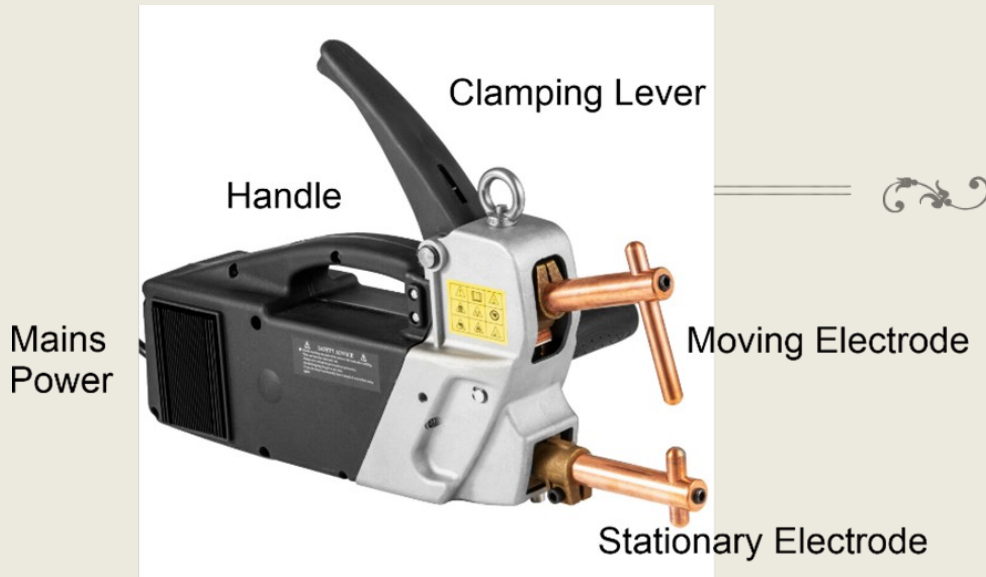


(j) Twisted

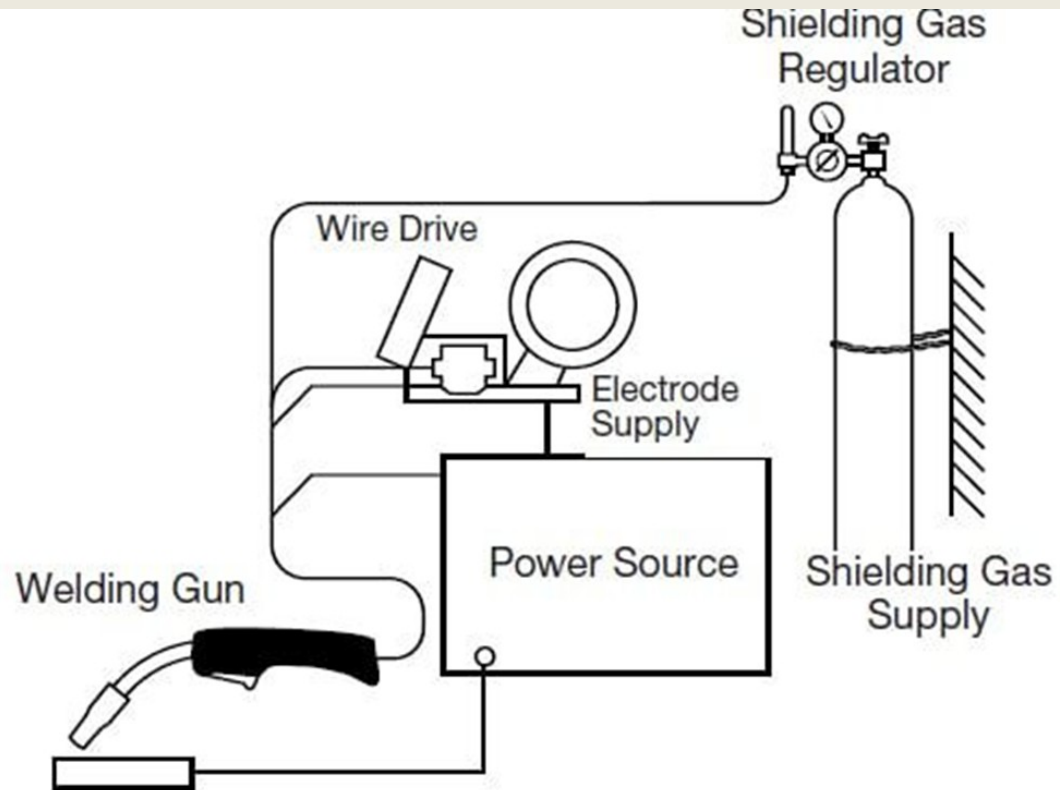
Stick Welder



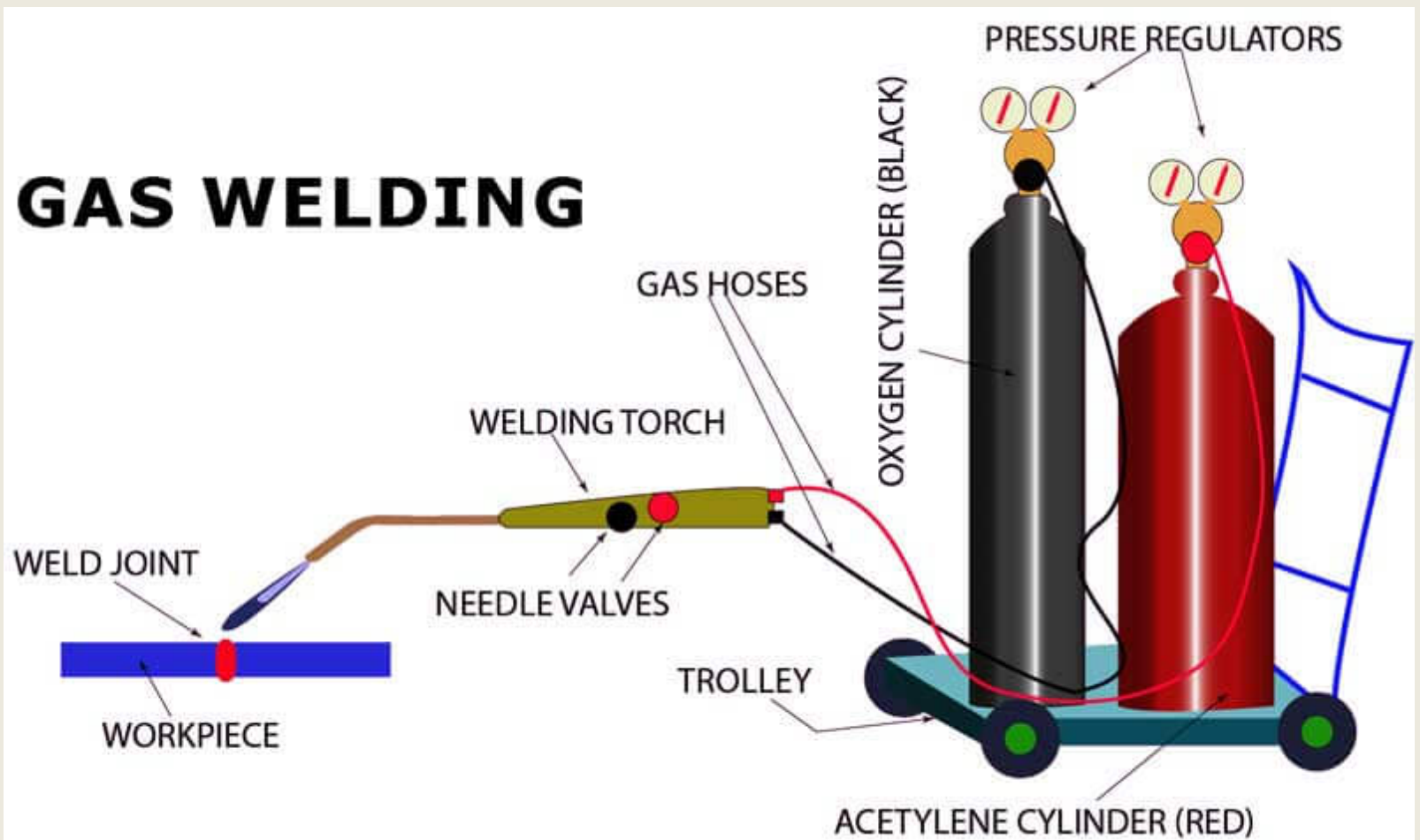
SPOT WELDER



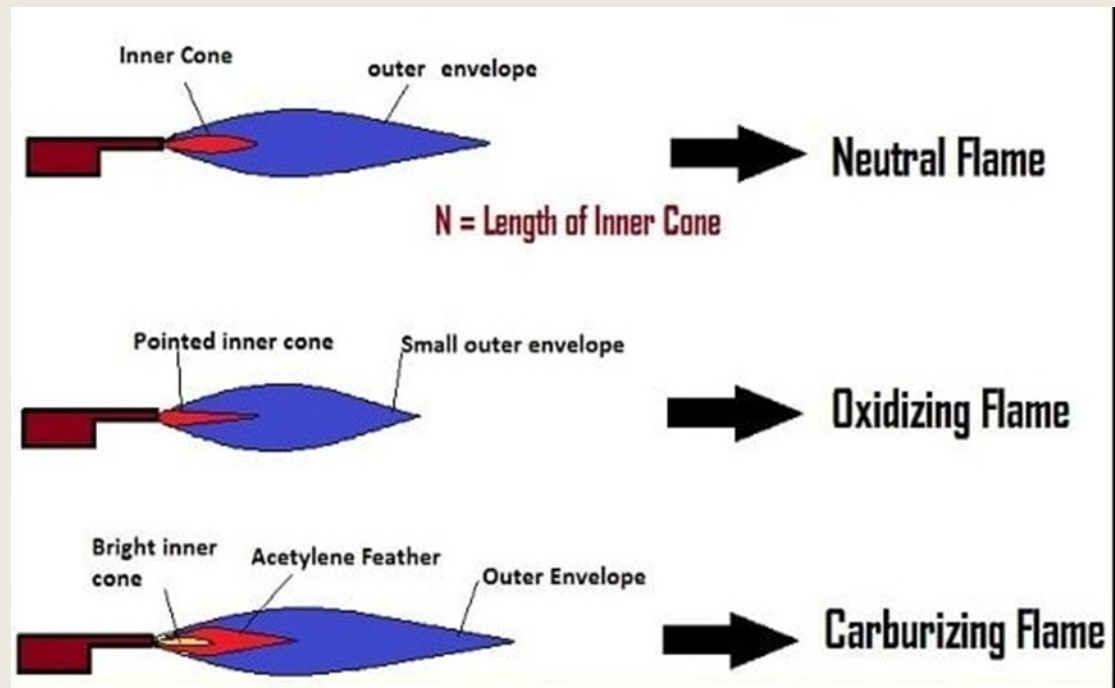
MIG Welder



Gas Welding



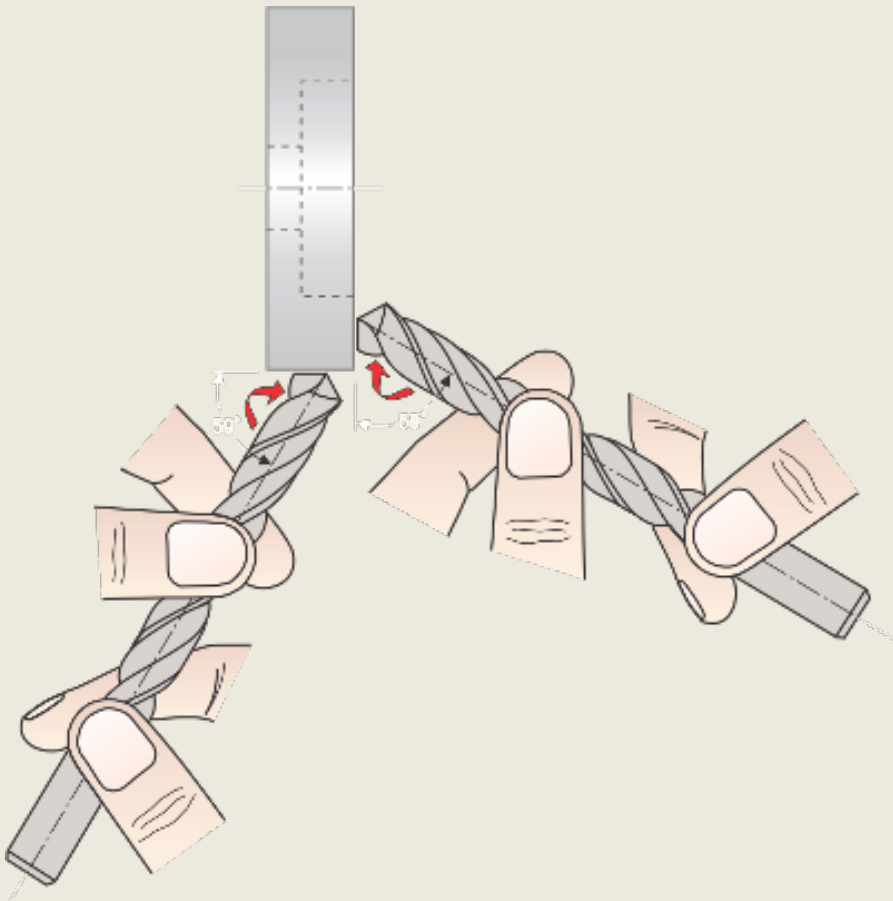
Gas Welding



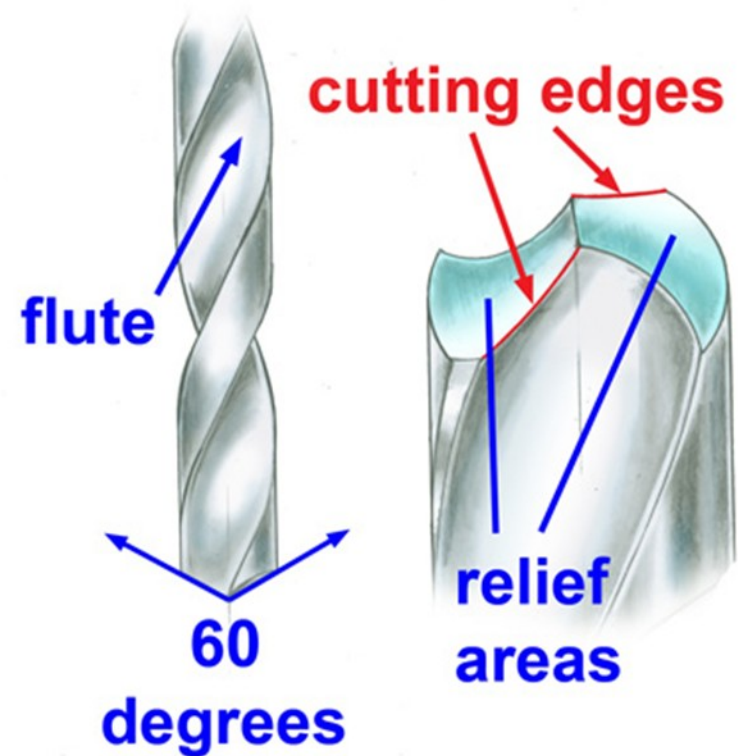
POWER DRILL



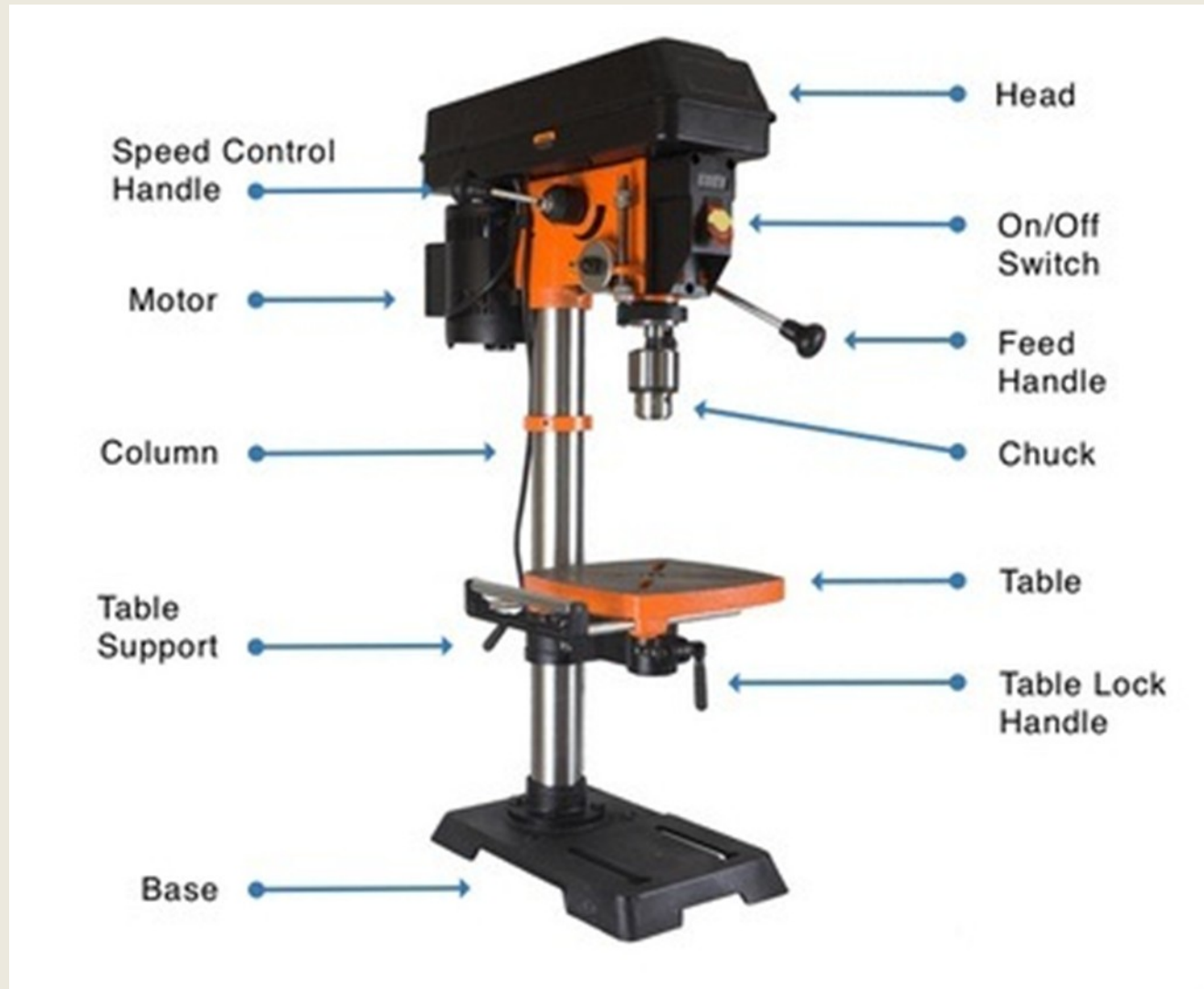
Sharpening Drill Bits



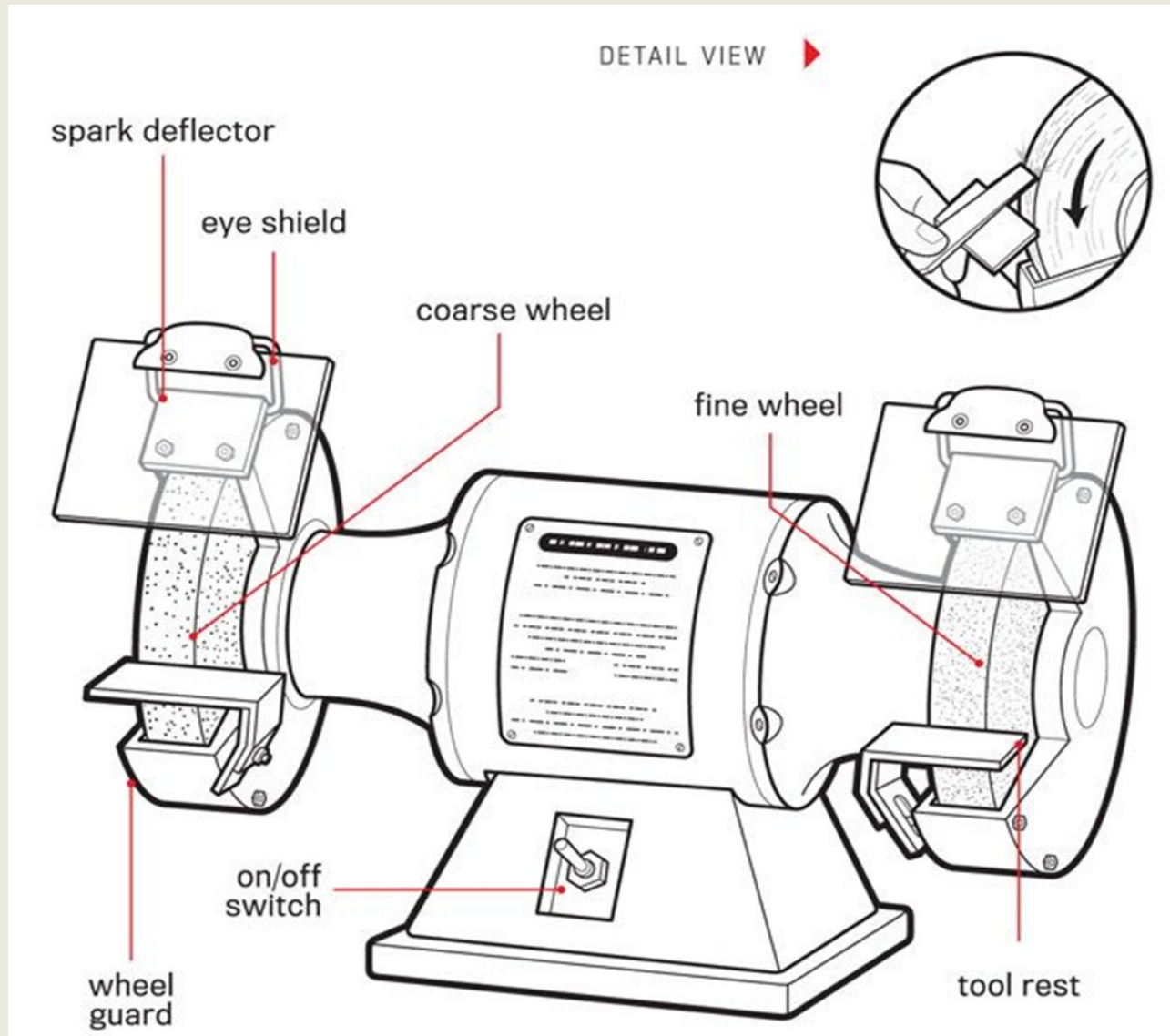
Parts of a Drill Bit



The Drill Stand

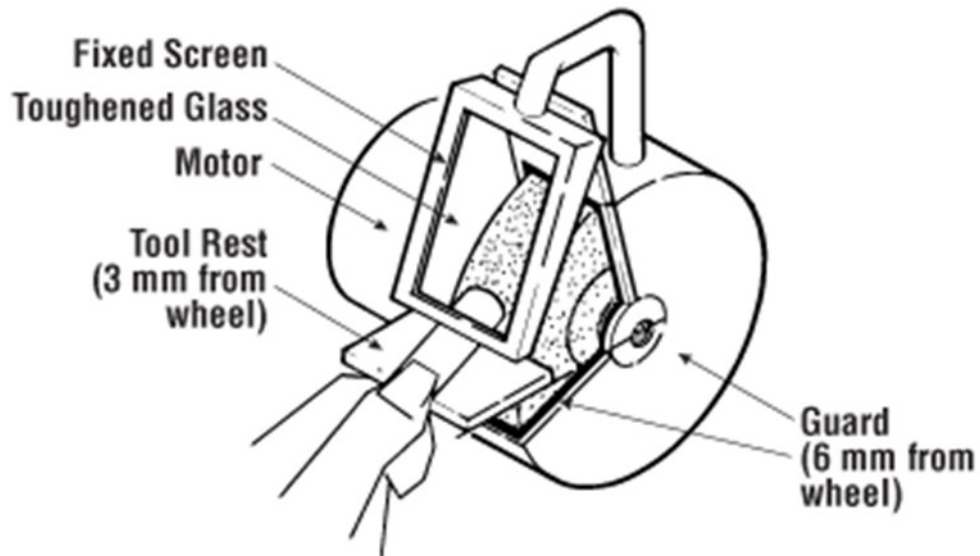


Bench Grinder





Bench Grinder



Angle Grinder

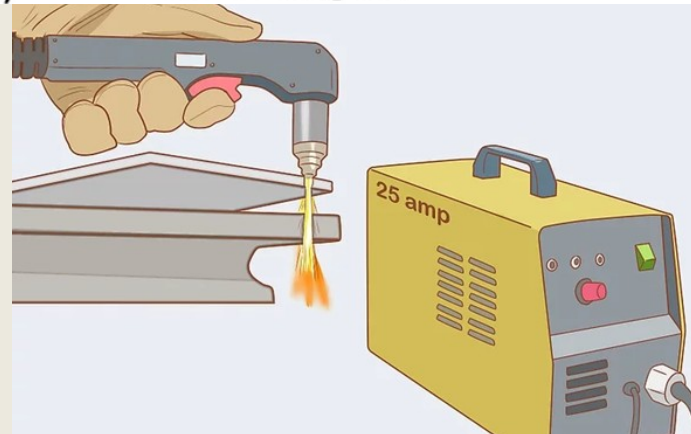


The diagram illustrates the TXI CUT 4060 P Plasma Cutter and its components. The main unit is shown in two views: a front view (left) and a side view (right). The front view shows the current control panel (1), access panel (2), torch cable (3), ground clamp (10), and eye protection glass (12). The side view shows the air regulator and filter (4), cooling fan (13), and torch head (5). A detailed view of the torch assembly is shown on the right, including the torch head (5), gas distributor (7), constricting nozzle (8), shielding cup (9), and copper electrode (6). A list of consumables is provided at the bottom left, including the copper electrode (6), gas distributor (7), constricting nozzle (8), and shielding cup (9).

1. Current control panel
2. Access panel
3. Torch cable
4. Air regulator and filter
5. Torch head
6. Copper electrode (cathode)
7. Gas distributor
8. Constricting nozzle
9. Shielding cup
10. Ground clamp
11. Brush
12. Eye protection glass
13. Cooling fan

Consumables

6. Copper electrode (cathode)
 7. Gas distributor
 8. Constricting nozzle
 9. Shielding cup



Shearing Machine